
Univariate Multiple Linear Regression

**Models
Parameters
Methods
Performance
Prediction
Generalizations**

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Introduction and intended audience

Regression analysis techniques are among the most often used statistical methods. Basically regression analysis involves fitting observations to an assumed model in an “optimal” way, using some predefined criterion for optimality or “best fit”. Afterwards various statistical inferences can be made on the model parameters and the model can be used for predicting new (or maybe hypothetical) observations unavailable at the time of model fitting.

Many different regression models and techniques of varying complexity exist. In this document the focus is on so called Univariate Multiple Linear Regression (UMLR). In this family of models, a measurable scalar quantity, Y , often called the response variable, is supposed to be a (linear) function of a set of controlled variables, $x_1, x_2, \dots, x_k$, often called predictor variables. As an example, the response variable may be a company’s energy consumption over some time period and the predictor variables may be various production specific quantities such as number of produced units of different sorts as well as e.g. meteorological data for the given time period.

In the following a relatively thorough description of the UMLR-model, its parameters and its use for prediction is given. Concise mathematical definitions and results are given followed by a detailed verbal explanation. Since the focus is on application of the method, derivations of results are omitted. In this way it is hoped that the document could be useful both for readers with an engineering or scientific background including a basic course on statistics and also for readers with only a very brief exposure to probability theory and statistics. Assumed is, however, at least informal understanding of concepts like random variables, mean value, variance, standard deviation, probability distributions in general and the normal (Gaussian) distribution in particular.

PART A: Model

1 Basic UMLR model

As stated above, the UMLR model in its basic form assumes that the scalar response variable, Y , is a linear function of a set of k predictor variables, $x_1, x_2, \dots, x_k$

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

The $k + 1$ model (regression) coefficients, $\beta_0, \beta_1, \dots, \beta_k$, are assumed to be unknown but constant quantities.

It can be seen that the basic model is linear not only in the predictor variables but also in the coefficients, the latter point proving to be most important for the following.

Usually the above equation does not explain everything about the variation of the response variable. To model deviations from the perfect fit and perhaps also to include the possibility of measurement errors in Y , an error or residual term, e , is added to the model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + e$$

The error term is assumed to be a random variable with zero mean and constant variance

$$E[e] = 0 \quad V[e] = \sigma^2$$

The predictor variables are assumed to be controlled, deterministic quantities (not random variables). However, due to the error term, the response variable will also be a random variable with mean and variance

$$E[Y|x_1, x_2, \dots, x_k] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k \quad V[Y|x_1, x_2, \dots, x_k] = \sigma^2$$

The purpose of the regression analysis is then based on observations to

- make estimates, $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ of the unknown coefficients, $\beta_0, \beta_1, \dots, \beta_k$
- make an estimate, $\hat{\sigma}^2 = s^2$, of the error term variance, σ^2
- make inferences on model parameters
- use model for prediction

For $k = 1$ the model simplifies to Simple Linear Regression which is the well-known problem of fitting a straight line to observation points in the two-dimensional (x,y)-plane. For $k = 2$ the problem is that of fitting a plane to points in the three-dimensional (x,y,z)-space, while in general the problem is that of fitting a k -dimensional hyperplane to points in a $(k + 1)$ -dimensional hyperspace.

Finally it should be noted that sometimes UMLR-models are made for several response variables, e.g. Y_1, Y_2, Y_3 , based on the same set of predictor variables, $x_1, x_2, \dots, x_k$. An example could be consumption of electricity, water and heat for producing a number of units of different sorts. In this context, however, these will be treated as separate, individual UMLR-models. A (somewhat more complex) alternative would be to join the response variables into a vector of simultaneous observations and then to use a Multivariate Multiple Linear Regression model. This allows for modeling of correlation between response variables and inter-experiment model errors.

2 Observations

The starting point for the regression analysis is gathering of n observations of the response variable corresponding to n combinations of the k predictor variables

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + e_i, \quad i = 1, 2, \dots, n$$

Note that in this equation, only Y_i and $x_{i1}, x_{i2}, \dots, x_{ik}$ are observable, while $\beta_0, \beta_1, \dots, \beta_k$ and e_i are non-observable and are to be estimated in the following.

Equations and computer implementations are greatly facilitated by introducing vector and matrix notation, therefore, defining vectors for coefficients and for observation i as

$$\begin{aligned} \boldsymbol{\beta} &= [\beta_0 \ \beta_1 \ \dots \ \beta_k]^T \\ \mathbf{x}_i &= [1 \ x_{i1} \ \dots \ x_{ik}] \end{aligned}$$

the above equation can be simplified to

$$Y_i = \mathbf{x}_i \boldsymbol{\beta} + e_i, \quad i = 1, 2, \dots, n$$

Furthermore, collecting terms for all n observations yields

$$\begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix}$$

Finally, defining

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nk} \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix}$$

the whole set of n simultaneous observations of the response variable and the k predictor variables can be written compactly as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

3 Assumptions

The main assumption is the basic UMLR model itself as stated above.

The n error terms $\{e_i\}_{i=1,\dots,n}$ are assumed to be independent random variables with zero mean and constant variance

$$E[e_i] = 0 \quad V[e_i] = \sigma^2$$

These are the only necessary assumptions for estimating the unknown regression coefficients. However, for estimating the error variance and for making inferences on regression coefficients, e.g. to test which predictor variables have significant influence on the response variable, it is necessary also to assume a distribution of the error random variables. Usually the error terms are assumed to be independent Gaussian random variables with constant variance

$$e_i \sim N(0, \sigma^2), \quad i = 1, 2, \dots, n$$

As a consequence of these assumptions, the n response variable observations $\{Y_i\}_{i=1,\dots,n}$ will also be independent Gaussian random variables with mean value as claimed in the linear model and with constant variance

$$\begin{aligned} Y_i &\sim N(x_i\boldsymbol{\beta}, \sigma^2), \quad i = 1, 2, \dots, n \\ E[Y_i] &= x_i\boldsymbol{\beta} \\ V[Y_i] &= \sigma^2 \end{aligned}$$

PART B: Estimation of model parameters

4 Estimation of regression model coefficients

Having collected the n observations, the next step is fitting the observations to an assumed model in an “optimal” way, using some predefined criterion for optimality or “best fit”. Several criteria, or cost functions, can be used for this purpose. However, the usual approach is to use a Least Squares criteria, i.e. minimizing the sum of the squared estimated residuals. This is mainly because this approach gives a nice analytical solution to the regression estimation problem, as opposed to e.g. minimizing the sum of absolute values of the estimated residuals.

The Least Squares criteria, however, has the downside that it is very sensitive to observation outliers, since the contributions of these will dominate the error sum of squares, quantified by their *leverage* (which is a number that can be calculated pr. observation to show its relative influence on the estimation result).

Since the true error terms are unknown quantities (random variables), their estimates have to be used in the Least Squares minimization procedure. Assuming for a moment that the regression coefficients, collected in the vector $\hat{\beta}$, have already been determined. The point estimate for response variable observation i is then its mathematical expectation, which according to the UMLR model is

$$\hat{Y}_i = E[Y_i | x_i] = x_i \hat{\beta}, \quad i = 1, 2, \dots, n$$

The point estimate for error term i is then also its mathematical expectation, which is

$$\hat{e}_i = Y_i - \hat{Y}_i = Y_i - x_i \hat{\beta}, \quad i = 1, 2, \dots, n$$

The Least Squares method then works by minimizing the total sum of squared estimated error terms, often called *SSE* (for Sum of Squares of Error terms). This sum can be expressed as

$$SSE = \sum_{i=1}^n \hat{e}_i^2 = \hat{\mathbf{e}}^T \hat{\mathbf{e}} = \sum_{i=1}^n (Y_i - x_i \hat{\beta})^2 = (\mathbf{Y} - \mathbf{X} \hat{\beta})^T (\mathbf{Y} - \mathbf{X} \hat{\beta})$$

Minimizing SSE (formally by taking partial derivatives, equating to zero and solving) gives the point estimates for the $k + 1$ UMLR model regression coefficients

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

This point estimate has the important consequence that the estimated (fitted) k -dimensional hyperplane will always go through the “center of gravity” point $(\bar{x}, \bar{Y})$ in the $(k + 1)$ -dimensional hyperspace.

Another property is that the estimated error terms sum to zero.

When the Gaussian assumption is included (not used yet), the Least Squares estimate of the regression coefficients is identical to the Maximum Likelihood estimate.

Finally, note that the above Least Squares criteria use the one-dimensional distances (“vertical” distances) between observations and their estimates. Sometimes it is desirable instead to use the $(k + 1)$ -dimensional Euclidian distance leading to a (more complicated) Total Least Squares method.

As a practical commentary can be noted that if the predictor variables are highly correlated, the matrix inversion in the above equation is numerically unstable (close to singular) and should then be calculated using so called SVD, Singular Value Decomposition.

So far, only the point estimates for the regression coefficients have been determined. To make inferences on them, e.g. to test which of them are significantly different from zero, or equivalently, which predictor variables have significant influence on the response variable, it is necessary to include the above mentioned assumption, that the error terms and consequently the response variable observations are both independent Gaussian random variables with constant variance.

Since the vector of estimated regression coefficients, $\hat{\beta}$, as seen above, is a function of the observation random variables collected in vector Y , the estimated regression coefficients are random variables themselves. Moreover, since each element in $\hat{\beta}$ is a linear combination of the elements of Y , which is a vector of independent Gaussian random variables, the vector of estimated regression coefficients, $\hat{\beta}$, has as simultaneous distribution a so called Multivariate Normal (Gaussian) distribution, or formally

$$\hat{\beta} \sim MVN(\beta, \Sigma_{\hat{\beta}})$$

This shows that the estimator $\hat{\beta}$ for the true unknown coefficients β is unbiased, i.e. its expected value is correct

$$E[\hat{\beta}] = \beta$$

The variance-covariance matrix for the estimator $\hat{\beta}$ can be calculated as

$$\Sigma_{\hat{\beta}} \stackrel{\text{def}}{=} \begin{bmatrix} V[\hat{\beta}_0] & Cov[\hat{\beta}_0, \hat{\beta}_1] & \dots & Cov[\hat{\beta}_0, \hat{\beta}_k] \\ Cov[\hat{\beta}_0, \hat{\beta}_1] & V[\hat{\beta}_1] & & \vdots \\ \vdots & \vdots & & \vdots \\ Cov[\hat{\beta}_0, \hat{\beta}_k] & \dots & \dots & V[\hat{\beta}_k] \end{bmatrix} = \sigma^2 (X^T X)^{-1}$$

In general the elements $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ of $\hat{\boldsymbol{\beta}}$ are correlated, i.e. the covariance elements in $\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}$ are non-zero. For the Simple Linear Regression case, i.e. for $k = 1$, this is easy to see: if the y-axis intercept coefficient $\hat{\beta}_0$ is increased, then the slope coefficient $\hat{\beta}_1$ must decrease for a given set of (x,Y) observation points, thus in this case $Cov[\hat{\beta}_0, \hat{\beta}_1] < 0$. This argument can be generalized to the cases $k > 1$.

The simultaneous distribution of $\hat{\boldsymbol{\beta}}$ and its mean and variance-covariance matrix can be used for making simultaneous inference on the regression coefficients.

If only marginal inferences are to be made on each coefficient one at a time disregarding the others, only the marginal distributions of the coefficients are needed.

Since the simultaneous distribution of $\hat{\boldsymbol{\beta}}$ is Multivariate Gaussian, it follows directly that the marginal distribution for each of the coefficients is the usual one-dimensional Gaussian distribution, i.e.

$$\begin{aligned}\hat{\beta}_j &\sim N(\beta_j, \sigma_{jj}^2), \quad j = 0, 1, \dots, k \\ E[\hat{\beta}_j] &= \beta_j \\ V[\hat{\beta}_j] &= \sigma_{jj}^2 = \{\sigma^2(\mathbf{X}^T \mathbf{X})^{-1}\}_{jj}\end{aligned}$$

5 Estimation of regression model variance

The variance of the UMLR model error terms, σ^2 , is also needed for making inferences on regression coefficients. Since its true value is unknown, it must be estimated. For that purpose, the above defined sum of squares of errors, SSE, can be used. SSE is a (quadratic) function of response variable observations, Y_i , and is thus a random variable itself. Actually it turns out that it is proportional to a so called chi-square random variable, or more precisely

$$\frac{SSE}{\sigma^2} = \frac{\sum_{i=1}^n (Y_i - \mathbf{x}_i \hat{\boldsymbol{\beta}})^2}{\sigma^2} \sim \chi_{n-(k+1)}^2$$

The subscript for the chi-square random variable symbol indicates the number of degrees of freedom, which is the distributions only parameter.

From this it follows that an unbiased point estimator for σ^2 is

$$\begin{aligned}\hat{\sigma}^2 = s^2 &= \frac{SSE}{n - (k + 1)} \sim \frac{\sigma^2}{n - (k + 1)} \chi_{n-(k+1)}^2 \\ E[\hat{\sigma}^2] &= \sigma^2 \\ V[\hat{\sigma}^2] &= \frac{2\sigma^4}{n-(k+1)}\end{aligned}$$

Furthermore, the point estimators $\hat{\sigma}^2$ and $\hat{\boldsymbol{\beta}}$ are independent.

PART C: Model control

6 Analysis of residuals

Having estimated the unknown parameters of the UMLR-model, various checks of the validity of the model should always be performed before using it to make inferences and predictions. These checks include graphical as well as quantitative inspections. This will be described in this and the following two sections. Here the term “model control” means validation of the UMLR-model with the pre-selected number of predictor variables. Even if this chosen UMLR-model might be validated following the guidelines sketched below, it might turn out that an even better UMLR-model with either fewer or more (e.g. polynomial, interaction or other non-linear) predictor variables can be found. These points will, however, be addressed in the sections on model simplifications and model extensions, respectively.

For Simple Linear Regression ($k=1$) it is easy visually to check the validity of the model by simply making a scatter plot of the observations of the response variable against the predictor variable together with a plot of the estimated linear estimate. Already for two predictor variables it becomes hard visually to check the fit of the estimated regression plane and for more than two predictor variables it becomes impossible.

The estimated residuals, however, are always one-dimensional no matter how many predictor variables are used in the model. The residuals are, as stated above, given as the difference between response variable observations and their estimates

$$\hat{e}_i = Y_i - \hat{Y}_i = Y_i - \mathbf{x}_i \hat{\boldsymbol{\beta}}, \quad i = 1, 2, \dots, n$$

For convenience, sometimes the estimated residuals are standardized by dividing them by the estimated error standard deviation

$$\hat{e}_{i,s} = \hat{e}_i / \hat{\sigma} = (Y_i - \hat{Y}_i) / \hat{\sigma} = (\mathbf{x}_i \hat{\boldsymbol{\beta}}) / \hat{\sigma}, \quad i = 1, 2, \dots, n$$

Recall that the assumptions on the error or residual terms are:

The n error terms $\{e_i\}_{i=1,\dots,n}$ are assumed to be independent Gaussian random variables with zero mean and constant variance

$$e_i \sim N(0, \sigma^2), \quad i = 1, 2, \dots, n$$

Zero mean error terms

Since the regression coefficients estimators are unbiased, this assumption is automatically achieved for the residuals

$$E[\hat{e}_i] = E[Y_i - \hat{Y}_i] = \mathbf{x}_i \boldsymbol{\beta} - E[\mathbf{x}_i \hat{\boldsymbol{\beta}}] = \mathbf{x}_i \boldsymbol{\beta} - \mathbf{x}_i \boldsymbol{\beta} = 0, \quad i = 1, 2, \dots, n$$

Independent error terms

This is the most important assumption for use of the UMLR model. It can visually be checked by plotting the residuals $\hat{e}_i$ (or the standardized residuals $\hat{e}_{i,s}$) as a function of the observation index $i = 1, 2, \dots, n$. This plot should ideally show a “randomly varying” pattern, that is, one without trends or systematic variation. More advanced analysis can also be performed by considering the residual terms a time series (or a “signal”) and calculating autocorrelation and power spectral density functions. If the residuals are true random (“white noise”) the autocorrelation plot should show an approximate zero-lag impulse and the power spectrum should be approximate flat.

If the independency assumption does not seem to be reasonable fulfilled, this is often an indication of an un-fitting model. Several reasons for this are possible: there may be missing predictor variables in the model, non-linear terms may be missing or perhaps interaction terms between predictor variables are missed. This should always lead to a renewed model specification until the independency property is approximately achieved.

Constant variance error terms

The second most important assumption is that the variance of the error terms does not vary with index number or with the value of response and predictor variable values. An indication of whether this is fulfilled can already be inspected on the above described plot. It is, however, useful also to plot the residuals $\hat{e}_i$ against the estimated response variable $\hat{Y}_i$ and optionally also against the predictor variables $x_1, x_2, \dots, x_k$ one at a time. These plots should ideally all show a random pattern of “constant vertical variation height”.

A typical deviation from the ideal is that the variances of the residuals increase with the response variable, which would be apparent in the plots if the “variation height” seems to have an (perhaps linear) increasing trend. If this tendency is very marked and there are very few observations, e.g. $n < 20 - 30$, a so called Weighted Least Squares approach can be used. This works by weighting each element of the Sum of Squares minimized in the Least Squares procedure according to the variance (high variance should give small weight and vice versa). More details on Weighted Least Squares will be given in a later section.

A non-constant variance of error terms could also be an indication of a non-fitting model as described above.

Gaussian error terms

This is the least important of the assumptions, that is, an approximate fulfillment is acceptable for use of the regression model. The first inspection is to plot a histogram of the residuals, that is, plot the relative frequency of residual terms values in appropriate intervals. Ideally this should show a symmetric, bell-shaped distribution around mean value zero, but slight deviations from this is acceptable. Optionally the histogram can be complemented by making a so called Normal Distribution Quantile-Quantile plot (QQ-plot). If the estimated residuals are distributed approximately Gaussian, they should result in an approximate straight line on the QQ-plot.

Hypothesis test of assumptions

For completeness it should be noted that formal statistical hypothesis tests are actually available for testing independency, constant variance and Gaussian distribution of the residuals. This point will, however, not be pursued here, since the above suggested plots are usually sufficient.

Outliers

If some of the observations clearly deviate from their estimated values based on the derived regression model, their respective residuals should appear in the above suggested plots as outliers. Outlier detection is almost a science in its own right and various schemes exist for detecting outliers, some of them involving the concept of leverages, which is a quantitative measure of the influence of the individual observations to the resulting regression estimates. This will, however, not be further described here. It is always good practice before just deleting outlying observations, to try to find a reason for their un-fitting values, and also to compare the regression results with and without the outliers present in the data.

7 Decomposition of total variation (ANOVA)

Following the graphical model inspections described in the previous section, the next step is a quantitative validation of the regression model. This can be done using an approach called Analysis of Variances (ANOVA), which will be described in this and the following section.

The sample mean value of the response variable observations is

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

The total variation of the n individual Y_i values around $\bar{Y}$ can be decomposed in two parts

$$\begin{aligned} SST &= \sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \\ &= SSE + SSR \end{aligned}$$

Here, SSE is the previously defined Sum of Squares of estimated Errors (residuals), which describes the random variation of observations Y_i from their estimated (or regressed) values $\hat{Y}_i$.

Likewise, SSR , the Sum of Squares of Regression, describes the part of the total variation of observations around their mean value $\bar{Y}$, that can be explained by the regression model estimated values $\hat{Y}_i$.

Loosely speaking, if the regression model is well-fitting the observations, SST should be dominated by SSR , or equivalently $SSR \gg SSE$. This point is quantified by the so called coefficient of multiple determination, R^2 , sometimes just called "R-squared". It is defined

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}, \quad 0 \leq R^2 \leq 1$$

Thus, R^2 indicates the fraction of the total variation of response variable observations that can be explained by the assumed regression model. If the regressed model shows a perfect

fit to the data (a hypothetical situation) $R^2 = 1$. For more realistic situations, $R^2 < 1$, and the closer R^2 is to 1, the better the model fits the observations.

The coefficient of multiple correlation R is by definition the positive square root of R^2 . For Simple Linear Regression this is equivalent to the absolute value of the sample correlation coefficient, r .

$$R^2 = r^2 \quad |r| = \sqrt{R^2}$$

It is evident then, that R^2 is insensitive to whether the response variable is positively or negatively correlated with the predictor variables – as opposed to r , where the sign gives exactly this information.

R^2 is often used as a “quality” or “performance” indicator for the fit of the regression model. When considering only one UMLR-model with a specific set of selected predictor variables, this makes perfectly good sense, a value of $R^2 > \text{e.g. } 0.75 - 0.80$ indicating a relative good fit of the model.

When several different models are considered with different sets of predictor variables, however, care should be taken if R^2 -values are used for comparison of the models. This is because the fraction SSE/SST is not an unbiased estimator of the fraction between error and total variances, respectively. As a consequence of this, when adding more predictor variables to the model, SSR can never decrease and will typically increase. Since SST has a fixed value for a given set of observations, this means that R^2 can never decrease and will typically increase when adding supplementary predictor variables to the model. This gives a risk of “over-fitting” the model (using too many variables), since it is hard to evaluate how much R^2 should increase when adding new predictor variables, for these to have significant influence on the response variable observations.

This point is addressed by modifying R^2 to the so called “adjusted coefficient of multiple determination” defined by

$$R_{adj}^2 = 1 - \frac{SSE/df_{SSE}}{SST/df_{SST}} = 1 - \frac{SSE/(n - k - 1)}{SST/(n - 1)} = 1 - \left(\frac{n - 1}{n - k - 1} \right) \frac{SSE}{SST} \leq R^2$$

The modification amounts to normalizing the Sum of Squares by their degrees of freedom, a parameter that will be explained further in the next section. Unlike R^2 , R_{adj}^2 is an unbiased estimator of the mentioned variances ratio. This means that when adding new candidate predictor variables to the model, R_{adj}^2 can in fact decrease if the increase in SSR is “too small”, making R_{adj}^2 a more sound parameter for comparison of models than R^2 .

As stated in the equation R_{adj}^2 will always be smaller than or equal to R^2 . It can also be seen that the modification of R^2 to R_{adj}^2 is most important if the number of observations, n , is not much higher than the number of predictor variables, k . If however $n \gg k$, then $R_{adj}^2 \cong R^2$.

8 Test for significant overall regression model (ANOVA)

When the (adjusted) coefficient of multiple determination has been calculated and evaluated, the next step is often a formal hypothesis test as to whether the specified UMLR-model is in fact statistically significant, that is, if the part of the total variation explained by the model, significantly is above the level of random variations due to error terms.

This can be done by making an ANOVA-table as shown below and explained in the following.

Variation	Sum of squares	Degrees of freedom	Mean Square	Hypothesis for test H_0	Test-statistic F_{test}	p -value for test
Regression	SSR	k	MSR	$\forall \beta_j = 0$ $j = 1, \dots, k$	$\frac{MSR}{MSE}$	$P(F_{k,n-k-1} > F_{test})$
Residual	SSE	$n - k - 1$	MSE			
Total	SST	$n - 1$				

Since Y_i , $\bar{Y}$, $\hat{Y}_i$ as a consequence of the assumptions on the error terms all are Gaussian random variables the three Sums of Squares, SST , SSE , SSR will all be proportional to chi-square distributed random variables, in fact

$$\frac{SST}{\sigma^2} \sim \chi_{n-1}^2 \quad \frac{SSE}{\sigma^2} \sim \chi_{n-(k+1)}^2 \quad \frac{SSR}{\sigma^2} \sim \chi_k^2$$

Again, the subscripts are the parameters of the chi-square random variables, namely their degrees of freedom. These will equal the number of random terms in the Sums of Squares minus the number of linear constraints between the random terms, or equivalently the number of estimated parameters in each Sum of Square.

For SST n observations are used and only one parameter is estimated, namely $\bar{Y}$, giving $n - 1$ degrees of freedom.

For SSE n observations are used and the $k + 1$ parameters in $\boldsymbol{\beta}$ are estimated, giving $n - (k + 1)$ degrees of freedom.

For SSR no observations are used, only the $k + 1$ parameters in $\boldsymbol{\beta}$ except β_0 , giving k degrees of freedom.

This justifies the number of degrees of freedom column in the ANOVA-table.

The next column is the Mean Squares column, which are unbiased estimates of the regression and residual variances respectively. These are simply defined by normalizing the Sums of Squares by their degrees of freedom

$$MSR = \frac{SSR}{df_{SSR}} = \frac{SSR}{k} \quad MSE = \frac{SSE}{df_{SSE}} = \frac{SSE}{n - 1 - k}$$

Note that MSE is exactly the previously defined estimated regression model or error term variance

$$MSE = \hat{\sigma}^2$$

Next, the hypothesis to be tested is whether the regression model significantly explains the variation in the response variable. This can be done by comparing the Mean Square variance estimates just defined. If the regression model fits the observations well, MSR should be significantly larger than MSE .

A formal statement of the hypothesis is

$$\begin{aligned} H_0: & \beta_1 = \beta_2 = \dots \beta_k = 0 \\ H_1: & \exists j \in \{1, 2, \dots, k\}: \beta_j \neq 0 \end{aligned}$$

H_0 is the null-hypothesis which, by standard statistical hypothesis testing technique, usually is desired to be rejected. In this case the null-hypothesis says that no regression coefficient is significantly different from zero, or equivalently, that the regression model does not help significantly in explaining the variation in response variable observations.

H_1 is the alternative hypothesis which must be accepted if H_0 is rejected. Here the alternative hypothesis says that one or more of the regression coefficients are important (significantly different from zero).

Note that the intercept coefficient β_0 is not a part of this test, since it is a constant term, and the hypothesis test is only about *variations* in Y .

If, alternatively, it is also important to test whether the intercept coefficient is different from zero, this can be done by modifying the definitions of the Sums of Squares SST, SSE, SSR defined previously from being *mean-corrected* to instead being *un-corrected*. This approach is, however, not customary and will not be used here.

To carry out a statistical test is always needed a so called Test Statistic. It is always a random variable, whose statistical distribution must be known, assuming H_0 is true.

In this case the test statistic, here named F_{test} , as introduced above compares the unbiased variance estimates MSR and MSE simply by calculating their ratio

$$F_{test} = \frac{MSR}{MSE} \sim F_{k, n-k-1}$$

Since both MSR and MSE are chi-square random variables divided by their degrees of freedom, their ratio is by definition a so called F-distributed random variable with parameters k and $n - k - 1$ as indicated by the subscripts. Both the test statistic F_{test} (being a ratio between variances) and the F-distribution in general, can only take on positive values.

When F_{test} is calculated using the observations data, it should be evaluated for being “small” or “large”. If H_0 were to be true, the value should lie in an interval from zero to at most “somewhat above 1” – allowing for random variations. If, however, F_{test} is “too large”, H_0 must be rejected and H_1 accepted. Note that this specific hypothesis test is *one-sided*, i.e. only high values of F_{test} leads to null-hypothesis rejection. Many other tests, relevant for other cases, are *two-sided*, i.e. rejection is done in both tails of the test statistic distribution.

Of course, in an exact science, all the quotation marks just used must be replaced by an exact criteria for deciding on either H_0 or H_1 based on the data. For all statistical hypothesis tests there are two standard ways of doing this. The first is to use a pre-defined so called level of significance, often named α , the other instead using a so called p-value. Since the two methods are used interchangeably – also in various statistical computer packages - they will both be explained for the present test statistic F_{test} and then used in the sequel.

The level of significance approach to hypothesis test concluding is based on the concept of Type I and II errors. When deciding on either accepting or rejecting H_0 , two types of errors can occur. A type I error occurs if H_0 is rejected when in fact it was true, whereas a type II error occurs if H_0 is accepted when in fact it was false. Type I errors are considered most serious, due to the standard use of hypothesis test methodology, where often the purpose of the test is to gain new insight by rejecting an a priori null-hypothesis. Rejecting H_0 is thus a *strong* statement, which in applications afterwards is often used to approve a new drug, a new production method and so on. On the other hand, accepting H_0 is a *weak* statement, since it gives no new information, the a priori null-hypothesis is still to be believed and nothing has changed. Or in other words, making a type I error is equivalent to erroneously change belief, while a type II error is equivalent to erroneously change nothing. How much this distinction of error types is relevant is clearly dependent on the concrete application, but the customary use of the concepts is as here explained.

Now, the level of significance concept is based on controlling the risk of making type I errors. Or more precisely

- Choose level of significance as the (small) fraction α , or equivalently $100 \cdot \alpha \%$
 - The criteria for either accepting or rejecting H_0 is then selected so that, given H_0 is in fact true, the probability (or risk) of making a type I error is exactly equal to α
- $$P(\text{type I error}) = P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$$

Depending on application typical values of α are 0.01, 0.05 or 0.10 (1, 5 or 10 %) where the first value is primarily used in applications where huge values or human lives are at stake, e.g. when approving a new drug, etc.

For the test at hand, the test of significant regression model, the procedure can then be sketched

- Choose level of significance as the (small) fraction α , or equivalently $100 \cdot \alpha$ %
- Calculate the test statistic value F_{test}
- If $F_{test} > F_{k,n-k-1,\alpha}$ reject H_0
- If $F_{test} \leq F_{k,n-k-1,\alpha}$ accept H_0

Here $F_{k,n-k-1,\alpha}$ is the upper $100(1 - \alpha)\%$ quantile of the F-distribution with parameters k and $n - k - 1$, or equivalently the value of the F-random variable where exactly α % of the probability mass is above and exactly $100(1 - \alpha)\%$ is below.

This procedure guarantees precisely, that even if $F_{test} > 1$, indicating that it is likely that not all regression coefficients are zero, this will only be used to in fact reject H_0 (and thereby significantly accepting that one or more coefficients are not zero), if the observed value of F_{test} is among the α % most rare of its possible upper values (one-sided test), given that H_0 were true. Thus it guarantees that $P(\text{type I error}) = \alpha$.

The quantile value $F_{k,n-k-1,\alpha}$ is a known value for the F-distribution and its computation is included in statistical computer packages.

Next, for the alternative p-value approach to hypothesis test concluding, first note that using the level of significance concept, the analysts message to the user is simply “Accept (or reject) H_0 at the α % level of significance” (hopefully also reporting the used number of observations, n).

More information can be given to the test conclusion user by not letting the analyst choosing at large a given level of significance α . Anyway, how should he choose it ? Instead the analyst can report a so called p -value for the test result.

The p -value can be defined in various equivalently ways. One is the following:

- If H_0 were to be true, the p -value is the probability that the test statistic (who's distribution as mentioned above is known, given H_0) should take on a value that is as “rare” or even “more rare” than its observed value. Here “rare” means that it should lie in the tail/tails of the distribution.

For the test at hand, the one-sided F-test of significant regression model, the procedure can then be sketched (as indicated in the ANOVA-table)

- Calculate the test statistic value F_{test}
- Then $p\text{-value} = P(F_{k,n-k-1} > F_{test})$

In this way, the p -value could be called a “rareness-indicator” since, if it has a very small value, the probability of H_0 being true is very small, given the observations leading to the value of the test statistic.

An equivalent definition of the p -value is to call it the “test-critical level of significance”. This is because for all values of $\alpha > p$ -value, H_0 should be rejected. As a numerical example, if p -value = 0.03 and $\alpha = 0.05$ were chosen, H_0 should be rejected, since the value of the test statistic is “more rare” than the limit specified by α . Conversely, if p -value = 0.03 and $\alpha = 0.01$ were chosen, H_0 should be accepted, since the value of the test statistic is “not rare enough” for rejection with the limit specified by α .

Using the p -value concept, the analysts message to the user amounts to simply reporting the p -value for hypothesis H_0 (again, hopefully also reporting the used number of observations, n). Then it is up to the user to decide if this should mean acceptance or rejection of H_0 given the application at hand. If the user actually is capable of using this extra information (as compared to just reporting accept/reject) is of course another discussion.

For completeness it should be mentioned that neither the level of significance approach nor the p -value approach give any control over type II errors. The risk of these errors can, however, also be controlled using the concept of *power functions* for hypothesis tests. Since this subject is a little more involved and customarily not used in standard hypothesis testing, it will not be pursued any further here.

A concluding remark on the test of significant regression model will be the following: If, using either the level of significance method or the p -value method, H_0 is actually rejected, usually the analyst and the user are “happy” or “satisfied” since it is then significantly proven (and remember, that is the best that can be achieved with statistics in place of deductions), that the assumed regression model explains a significant amount of the variations in response variable observations. Thus the regression is “good” and may be used for further purposes such as prediction described in a later section.

This is, however, absolutely no guarantee that the tested UMLR-model is the best possible UMLR-model. Several improvements might be possible such as

- there might be missing predictor variables in the model
- there might be too many predictor variables in the model
- non-linear, e.g. polynomial, terms might be missing in the model
- interaction terms between predictor variables might be missing in the model

It should be clear, that some methodology for comparing, simplifying, and choosing between UMLR-models would be appropriate if it is desired not just to accept the first proposed model, that turns out to be significant by the above test procedures.

This point will – among others - be elaborated on further in the next sections.

PART D: Use of estimated model

9 Some important questions for the validated model

At times the main purpose of a regression analysis is estimating the parameters for an UMLR model with a predefined set of predictor variables. This was achieved in sections 4 and 5 above. Very often, however, some natural follow-up questions are raised about the estimated and validated model. Among these could be, e.g.

Questions concerning model parameters:

- Could the unknown but estimated parameters hypothetically be assumed to take on some particular specified values ?
- Could one or some of them hypothetically be zero indicating unimportant and insignificant predictor variables ?
- What would be a probable interval for the value of each parameter ?

Questions concerning possible model simplification:

- Could the validated model possibly be simplified and what would be a reasonably simple procedure for doing this ?

Questions concerning using the model to make predictions:

- What would be probable values for response variable observations for combinations of predictor variable values not yet observed ?
- What would be a probable interval for a single new observation ?
- What would be a probable interval for the expected mean of a new observation ?

These interrelated questions and others will be addressed in the following sections 10, 11 and 12, respectively. To answer the questions, however, the model parameter estimators (random variables) and their distributions are needed. Below is, for convenience, restated the results needed plus some new information on use of the estimated error term variance.

The unbiased point estimator for the $k + 1$ UMLR model regression coefficients and their joint (simultaneous) distribution is

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} = (X^T X)^{-1} X^T Y \sim MVN(\beta, \Sigma_{\hat{\beta}})$$

with variance-covariance matrix

$$\mathbf{\Sigma}_{\hat{\boldsymbol{\beta}}} \stackrel{\text{def}}{=} \begin{bmatrix} V[\hat{\beta}_0] & Cov[\hat{\beta}_0, \hat{\beta}_1] & \dots & Cov[\hat{\beta}_0, \hat{\beta}_k] \\ Cov[\hat{\beta}_0, \hat{\beta}_1] & V[\hat{\beta}_1] & & \vdots \\ \vdots & \vdots & & \vdots \\ Cov[\hat{\beta}_0, \hat{\beta}_k] & \dots & \dots & V[\hat{\beta}_k] \end{bmatrix} = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

Note that even though the regression model and the response variable Y is univariate (scalar values), the distribution of the regression coefficients vector $\hat{\boldsymbol{\beta}}$ is truly multivariate, since in general the elements $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ of $\hat{\boldsymbol{\beta}}$ are correlated, i.e. the covariance elements in $\mathbf{\Sigma}_{\hat{\boldsymbol{\beta}}}$ are non-zero. As stated earlier this is even the case for Simple Linear Regression with only an intercept parameter and a slope parameter.

The marginal distributions of the elements $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ of $\hat{\boldsymbol{\beta}}$, which are relevant when the elements are addressed one-at-a-time disregarding the others, are

$$\begin{aligned} \hat{\beta}_j &\sim N(\beta_j, \sigma_{jj}^2), \quad j = 0, 1, \dots, k \\ E[\hat{\beta}_j] &= \beta_j \\ V[\hat{\beta}_j] &= \sigma_{jj}^2 = \{\sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}\}_{jj} \end{aligned}$$

where the subscripts jj in $\{\sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}\}_{jj}$ is indicating the diagonal element in $\mathbf{\Sigma}_{\hat{\boldsymbol{\beta}}}$ corresponding to coefficient β_j .

The unbiased point estimator for the UMLR model error term variance (or “noise”), σ^2 , and its distribution is

$$\begin{aligned} \hat{\sigma}^2 = s^2 &= \frac{SSE}{n - (k + 1)} \sim \frac{\sigma^2}{n - (k + 1)} \chi_{n-(k+1)}^2 \\ E[\hat{\sigma}^2] &= \sigma^2 \\ V[\hat{\sigma}^2] &= \frac{2\sigma^4}{n-(k+1)} \end{aligned}$$

Furthermore, the point estimators $\hat{\sigma}^2$ and $\hat{\boldsymbol{\beta}}$ are independent.

As can be seen above, the variance-covariance matrix $\mathbf{\Sigma}_{\hat{\boldsymbol{\beta}}}$ is a function of the unknown error term variance, σ^2 . Thus, to make the variance-covariance matrix and the marginal variances computable based on observations, it is necessary to substitute σ^2 with its unbiased estimate $\hat{\sigma}^2 = s^2$ giving

$$\mathbf{S}_{\hat{\boldsymbol{\beta}}} = \hat{\mathbf{\Sigma}}_{\hat{\boldsymbol{\beta}}} \stackrel{\text{def}}{=} \begin{bmatrix} \hat{V}[\hat{\beta}_0] & \widehat{Cov}[\hat{\beta}_0, \hat{\beta}_1] & \dots & \widehat{Cov}[\hat{\beta}_0, \hat{\beta}_k] \\ \widehat{Cov}[\hat{\beta}_0, \hat{\beta}_1] & \hat{V}[\hat{\beta}_1] & & \vdots \\ \vdots & \vdots & & \vdots \\ \widehat{Cov}[\hat{\beta}_0, \hat{\beta}_k] & \dots & \dots & \hat{V}[\hat{\beta}_k] \end{bmatrix} = \hat{\sigma}^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

and

$$s_{\hat{\beta}_j}^2 = \hat{V}[\hat{\beta}_j] = \hat{\sigma}_{jj}^2 = \{\hat{\sigma}^2 (\mathbf{X}^T \mathbf{X})^{-1}\}_{jj}$$

10 The regression table and marginal tests for predictor variables

The simplest approach for making inferences on regression coefficients is to do it in a univariate setting. This is also the usual method in various statistical computer packages and is equivalent to making marginal inferences on the coefficients, i.e. to address them one-at-a-time while disregarding the others. This approach will be used at this point, while methods for doing it in a multivariate setting, and thus making simultaneous inferences on several or all coefficients, will be deferred to a later section.

The marginal tests on regression coefficients are often reported in a Regression-table as shown below and explained in the following.

Predictor variable	Coefficient estimate	Estimated standard error for coefficient	Hypothesis for test H_0	Test-statistic t_{test}	p -value for test	Lower limit for confidence interval	Upper limit for confidence interval
intercept	$\hat{\beta}_0$	$S_{\hat{\beta}_0}$	$\beta_0 = 0$	$\hat{\beta}_0/S_{\hat{\beta}_0}$	$2P(t_{n-k-1} > t_{test,\beta_0})$	$CI_{\beta_0,low,\alpha}$	$CI_{\beta_0,high,\alpha}$
x_1	$\hat{\beta}_1$	$S_{\hat{\beta}_1}$	$\beta_1 = 0$	$\hat{\beta}_1/S_{\hat{\beta}_1}$	$2P(t_{n-k-1} > t_{test,\beta_1})$	$CI_{\beta_1,low,\alpha}$	$CI_{\beta_1,high,\alpha}$
...	...	...	...	...	...	...	...
x_k	$\hat{\beta}_k$	$S_{\hat{\beta}_k}$	$\beta_k = 0$	$\hat{\beta}_k/S_{\hat{\beta}_k}$	$2P(t_{n-k-1} > t_{test,\beta_k})$	$CI_{\beta_k,low,\alpha}$	$CI_{\beta_k,high,\alpha}$

To test if a selected regression coefficient β_j , $j = 0, 1, \dots, k$ based on the observations could be hypothesized to take on a particular value, say b_j , or formally

$$H_0: \beta_j = b_j$$

$$H_1: \beta_j \neq b_j$$

can be used the fact that the standardized estimator, $\hat{\beta}_j$, for β_j is a standard Gaussian random variable

$$\frac{\hat{\beta}_j - \beta_j}{\sigma_{\hat{\beta}_j}} \stackrel{H_0}{\underset{\sim}{=}} \frac{\hat{\beta}_j - b_j}{\sigma_{\hat{\beta}_j}} \sim N(0,1)$$

and the fact that the estimator for the marginal variance in this equation is proportional to a chi-square random variable with $n - k - 1$ degrees of freedom

$$\frac{s_{\hat{\beta}_j}^2}{\sigma_{\hat{\beta}_j}^2} \sim \frac{\chi_{n-k-1}^2}{n-k-1}$$

to define a t -distributed test statistic with $n - k - 1$ degrees of freedom

$$t_{test,\beta_j} \stackrel{\text{def}}{=} \frac{\hat{\beta}_j - b_j}{s_{\hat{\beta}_j}} \stackrel{H_0}{\underset{\sim}{\approx}} t_{n-k-1}$$

This test statistic can be used to make tests for any hypothesized value b_j for β_j . Here will be focused on testing the particular value $b_j = 0$, since this is equivalent to testing whether the matching predictor variable x_j (or the intercept term if the case $j = 0$ is considered) has significant influence on the response variable.

Thus the hypothesis is

$$H_0: \beta_j = 0$$

$$H_1: \beta_j \neq 0$$

and the test statistic is

$$t_{test, \beta_j} \stackrel{\text{def}}{=} \frac{\hat{\beta}_j}{s_{\hat{\beta}_j}} \stackrel{H_0}{\approx} t_{n-k-1}$$

as indicated in the regression table. It is clear, that either large positive or large negative values of the test statistic should lead to a rejection of the null-hypothesis, thus indicating that the predictor variable x_j is indeed significant.

To actually conduct the test for the present observation data, first the test statistic value is calculated. Then, as in any statistical hypothesis test, and as explained thoroughly for the test for an overall significant regression model in the previous section, the conclusion can be made using either the level of significance concept or the p -value concept.

To use the level of significance approach the procedure for this test is then

- Choose level of significance as the (small) fraction α , or equivalently $100 \cdot \alpha$ %
- Calculate the test statistic value t_{test, β_j} based on the observations data
- If $|t_{test, \beta_j}| > t_{n-k-1, \alpha/2}$ reject H_0
- If $|t_{test, \beta_j}| \leq t_{n-k-1, \alpha/2}$ accept H_0

Here $t_{n-k-1, \alpha/2}$ is the upper $100(1 - \alpha/2)\%$ quantile of the t -distribution with $n - k - 1$ degrees of freedom, or equivalently the value of the t -random variable where exactly $\alpha/2$ % of the probability mass is above and exactly $100(1 - \alpha/2)\%$ is below.

Note that this - as opposed to the test referred to previously - is a two-sided test, since the rejection values are in both ends of the random variable's range. This explains why the total probability mass α % that should lead to rejection of the null-hypothesis, is partitioned in two probability "tail" parts each of probability mass $\alpha/2$ %.

The quantile value $t_{n-k-1, \alpha/2}$ is a known value for the t -distribution and its computation is included in statistical computer packages.

Alternatively, to use the p -value approach for this two-sided test, the procedure is as indicated in the regression table

- Calculate the test statistic value F_{test}
- Then $p\text{-value} = P(|t_{n-k-1}| > |t_{test, \beta_j}|) = 2P(t_{n-k-1} > |t_{test, \beta_j}|)$

with the same interpretation of the p -value as explained previously.

Besides testing a hypothesis on a particular value (or zero) of a regression coefficient, it is often also of interest to state an interval in which the value of the unknown coefficient should lie with a “reasonable high probability”. This can be done by developing a so called Confidence Interval (CI) for the unknown coefficient.

To calculate a confidence interval, first a desired level of confidence must be selected, usually expressed in % as a $(1 - \alpha)100\%$ confidence interval, where typical values of α are 0.01, 0.05 or 0.10, giving 99%, 95% and 90% confidence intervals, respectively. The level of confidence expresses how “certain” the analyst is that the unknown coefficient actually lies in the given interval. Choosing this level of confidence is always a compromise since choosing a high level of confidence gives wide intervals while a lower level of confidence gives a more narrow interval.

Two remarks about confidence intervals are in place at this point.

First note that if the level of significance approach has been used for hypothesis testing as explained above, the same value of α can, but doesn’t have to, be used for the confidence interval as well.

Secondly, note the term “confidence” in the interval’s name. It stems from the fact that each time such an interval is actually calculated based on observations, either the true unknown value of the regression coefficient actually lies in the stated interval or it does not lie in the interval, and since the data are already collected, no probability can be calculated about them. The correct interpretation is therefore that the analyst has the reported amount of “confidence” in that it actually lies in the interval, or equivalently that if the data were collected and the interval calculated many times, then the fraction corresponding to the level of confidence of these intervals would in fact include the unknown coefficient. As long as a *frequency interpretation of probabilities* is kept in mind, however, there are no practical problems with the interpretation of the interval.

For the test at hand, to calculate a $(1 - \alpha)100\%$ confidence interval for the unknown value of regression coefficient β_j can be used the definition of quantiles for a (symmetric) t -distributed random variable with $n - k - 1$ degrees of freedom

$$P\left(-t_{n-k-1, \frac{\alpha}{2}} < t_{n-k-1} < t_{n-k-1, \frac{\alpha}{2}}\right) = 1 - \alpha$$

Then inserting the test statistic from above, which has exactly this t -distribution, gives

$$P\left(-t_{n-k-1, \frac{\alpha}{2}} < \frac{\hat{\beta}_j - \beta_j}{s_{\hat{\beta}_j}} < t_{n-k-1, \frac{\alpha}{2}}\right) = 1 - \alpha$$

Rearranging terms then gives

$$P\left(\hat{\beta}_j - s_{\hat{\beta}_j} \cdot t_{n-k-1, \frac{\alpha}{2}} < \beta_j < \hat{\beta}_j + s_{\hat{\beta}_j} \cdot t_{n-k-1, \frac{\alpha}{2}}\right) = 1 - \alpha$$

This is exactly the desired interval for the unknown regression coefficient β_j so, as stated in the regression table

$$(1 - \alpha)100\% \text{ confidence interval for } \beta_j: \left[CI_{\beta_j, low, \alpha}, CI_{\beta_j, high, \alpha}\right] = \left[\hat{\beta}_j \pm s_{\hat{\beta}_j} \cdot t_{n-k-1, \frac{\alpha}{2}}\right]$$

To finish this section, three comments will be given.

First, and perhaps most importantly, care should be taken when interpreting the results of the hypothesis tests and the confidence intervals developed in this section. As already stated at the start of the section, the approach is here to make marginal inferences and consider the predictor variables and their regression coefficients *one-at-a-time*. If several or all of the tests are considered *simultaneously*, then the specified level of significance or the calculated *p*-values are not valid for this simultaneous statistical statement for several coefficients. Likewise, if several or all confidence intervals are considered simultaneously, then the specified level of confidence is not achieved simultaneously. In both cases this is because, if simultaneous statements are desired, the problem should be addressed as a true multivariate statistical problem. Often, however, this point is ruthlessly ignored.

Secondly, for completeness, it should be mentioned that for the Simple Linear Regression case, the test devised for the regression coefficient β_1 (that is, the slope coefficient) in this section (a *t*-test) is equivalent to the test for overall significant regression model described in section 8 (a *F*-test), since the latter test ignores the intercept term, addressing only *variations* in the response variable. This is mathematically secured by the fact, that there is an intimate relationship between *t*- and *F*-distributed random variables.

Finally, tests and confidence intervals for the unknown error term variance, σ^2 , can also be devised along the same lines as described above for the unknown regression coefficients, the difference being the involved test statistic and its distribution. Also, inference for the variance is a true univariate statistical problem without the marginal-versus-simultaneous complications. Often, however, inference on the variance is omitted and will be here as well.

11 Model simplification procedure using marginal tests

Using the marginal tests for significant regression coefficients developed in the previous section, a simple method for screening of predictor variables can be devised. The suggested iterative procedure is a simple, reasonable, but not optimal way to examine, whether some of the pre-selected predictor variables in the model advantageously might be removed, leading to a simpler model.

Suggested iterative model simplification procedure starting with the pre-selected set of predictor variables:

1. Calculate estimated regression coefficients and variance as described in Part B, Estimation of model parameters
2. Calculate the ANOVA table and validate model as described in Part C, Model control
3. Calculate the Regression table as described in section 10
4. Find in the Regression table the regression coefficient marginal test with the highest p -value in the Regression table
5. If this p -value is above some pre-defined level, e.g. 0.05, remove the corresponding predictor variable and start anew at point 1
6. If the highest p -value in the Regression table is below the pre-defined level, e.g. 0.05, STOP

Note especially that after removing a predictor variable, the estimation procedure, the ANOVA table and the Regression should be calculated anew, since the model has now changed.

Care should be taken in step 4 if the two highest p -values in the regression table are relatively close, e.g. less than a factor 2-3 apart. If so, it might be tried first to remove one of the two involved predictor variables, afterwards removing the other, and then comparing the resulting regression models with either the one or the other removed, using the heuristic rating for the overall regression model described in section 13.

As stated above, this proposed procedure is not optimal in the sense, that it does not guarantee that the best of all possible regression models for the pre-defined set of predictor variables is found. However, for most practical cases, it will lead to a good regression model usable for predictions. Also, in the typical situation, it will lead to a model which is “almost as good” as the optimal, if comparing models using the adjusted R-square as performance indicator. This will be further addressed in sections 13 and 14.

If it is really important to be sure to achieve the optimal or a very-close-to-optimal model, either an exhaustive search of possible models, or a much more complex strategy using the Extra Sums of Squares concept can be used as described in Part E2. Both these approaches are, however, much more computationally complex than the above simple procedure, and also require more than just the Regression table’s marginal tests.

12 Use of regression model for prediction

Very often a regression model with estimated parameters is (also) used for making predictions, that is, forecasting response variable values for combinations of predictor variable values not included in the available observations. Sometimes only the expected value of the response variable is needed, but at times also confidence intervals for such values are desired. Below both these aspects will be addressed in turn.

To reintroduce the notation, the assumed model is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + e$$

The available observations are

$$(Y_i, \mathbf{x}_i) = (Y_i, [1 \ x_{i1} \ \dots \ x_{ik}]), \quad i = 1, 2, \dots, n$$

where the leading “1”-elements in the predictor variable vectors is for convenience of the vector notation.

These observations are used to calculate the Least Squares estimates of the regression coefficients

$$\hat{\boldsymbol{\beta}} = [\hat{\beta}_0 \ \hat{\beta}_1 \ \dots \ \hat{\beta}_k]^T$$

First, for making point estimate prediction for the response variable value, let the desired combination of predictor variable values be

$$\mathbf{x}_0 = [1 \ x_{01} \ \dots \ x_{0k}]$$

Thus, the response variable value to be predicted is $Y(\mathbf{x}_0)$.

Since the error term is supposed to be zero mean, i.e. $E[e] = 0$, the true expected value is

$$E[Y(\mathbf{x}_0)] = \mathbf{x}_0 \boldsymbol{\beta}$$

An obvious and unbiased point estimate predictor for this value is then achieved by substituting the coefficients vector by its estimate, giving

$$\hat{Y}(\mathbf{x}_0) = \mathbf{x}_0 \hat{\boldsymbol{\beta}}$$

This is, of course, relatively trivial, and is just the multiple regression generalization of the simple linear regression method of using the estimated line to make predictions.

Next, for prediction of confidence intervals for $Y(\mathbf{x}_0)$, the situation is a little more subtle. This is because, here it is necessary to distinguish between two types of confidence intervals for $Y(\mathbf{x}_0)$.

One type is a confidence interval for the expected mean of $Y(\mathbf{x}_0)$, i.e. a confidence interval for $E[Y(\mathbf{x}_0)] = \mathbf{x}_0\boldsymbol{\beta}$. This is to be interpreted as a confidence interval for the expected sample mean of $Y(\mathbf{x}_0)$ *if several observations of Y hypothetically could be made for $\mathbf{x} = \mathbf{x}_0$* . This kind of confidence interval is usually the relevant type, if it is actually not possible to make observations for $\mathbf{x} = \mathbf{x}_0$, but a probable interval for their mean Y -value is still of interest.

The second type is a confidence interval for $Y(\mathbf{x}_0)$ if actually the experimenter plans to do *one and only one real, new observation* of Y for $\mathbf{x} = \mathbf{x}_0$ and is a priori for some reason interested in a probable interval for its Y -value.

For calculation of the first type of confidence interval, a confidence interval for the expected mean $E[Y(\mathbf{x}_0)] = \mathbf{x}_0\boldsymbol{\beta}$ can be used the distribution result

$$\mathbf{x}_0\hat{\boldsymbol{\beta}} \sim N(\mathbf{x}_0\boldsymbol{\beta}, \sigma^2 \mathbf{x}_0(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0^T)$$

As used before, the unknown model error variance is substituted by its estimate, giving

$$\frac{\mathbf{x}_0\hat{\boldsymbol{\beta}} - \mathbf{x}_0\boldsymbol{\beta}}{\sqrt{\hat{\sigma}^2 \mathbf{x}_0(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0^T}} \sim t_{n-k-1}$$

Then, as also seen previously, using the quantiles of the t -distribution, gives the desired

$$(1 - \alpha)100\% \text{ confidence interval for the expected mean } E[Y(\mathbf{x}_0)] = \mathbf{x}_0\boldsymbol{\beta}: \\ \left[\mathbf{x}_0\hat{\boldsymbol{\beta}} \pm \sqrt{\hat{\sigma}^2 \mathbf{x}_0(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0^T} \cdot t_{n-k-1, \frac{\alpha}{2}} \right]$$

Next, for calculation of the second type of confidence interval, a confidence interval for a new observation $Y(\mathbf{x}_0)$ can be used the distribution result for a new observation

$$Y(\mathbf{x}_0) \sim N(\mathbf{x}_0\boldsymbol{\beta}, \sigma^2)$$

Using the distribution result for $\mathbf{x}_0\hat{\boldsymbol{\beta}}$ just stated gives

$$Y(\mathbf{x}_0) - \mathbf{x}_0\hat{\boldsymbol{\beta}} \sim N(0, \sigma^2 + \sigma^2 \mathbf{x}_0(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0^T)$$

Again, substituting the error variance by its estimate, gives

$$\frac{Y(x_0) - x_0 \hat{\beta}}{\sqrt{\hat{\sigma}^2(1 + x_0(X^T X)^{-1} x_0^T)}} \sim t_{n-k-1}$$

And finally, analogue to above, the desired interval is

$$(1 - \alpha)100\% \text{ confidence interval for a new observation } Y(x_0):$$

$$\left[x_0 \hat{\beta} \pm \sqrt{\hat{\sigma}^2(1 + x_0(X^T X)^{-1} x_0^T)} \cdot t_{n-k-1, \frac{\alpha}{2}} \right]$$

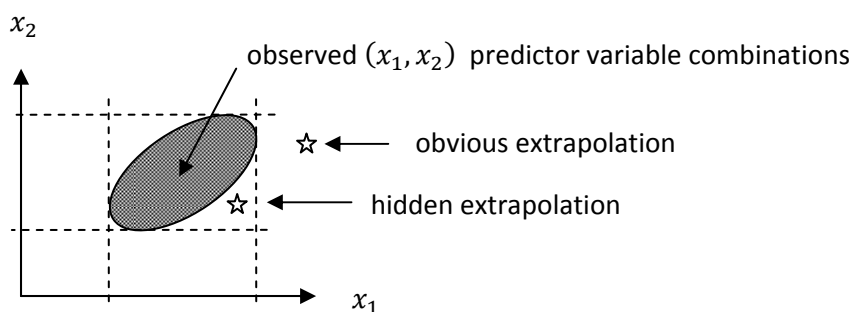
Note that this second type of confidence interval is always wider than the first type due to the extra “1” addend under the square root sign. The intuitive explanation to this is that a single new observation is more uncertain (has higher variance) than the expected mean of many such observations.

Finally in this section, a warning remark about using the estimated regression model to make predictions: as always, doing interpolations is more “safe” than doing extrapolations.

In the Simple Linear Regression is it simple to see if the new value of the single predictor variable is safely within its available values in the observations used to make the model parameter estimation. If not, that is if model extrapolations are made, the utmost care should be taken when predicting response variable values. This is well-known and trivial.

If there are more than one predictor variable, it is not equally trivial to see if the prediction being made is an extrapolation or not, since there can be hidden extrapolations. This is because, in the multidimensional case it is *not* enough to check that each predictor variable is well inside the interval covering its previously observed values, since this does not secure that the *simultaneous combinations* of several predictor variables are well inside their simultaneous values in the observations.

This is easy to see for just two predictor variables as sketched below, and generalizes obviously to more than two predictor variables. Methods for detecting hidden extrapolations do exist (some of them using the concepts *leverage* and the *hat matrix*), but will not be pursued any further here.



13 Overall heuristic rating of regression model

When is an estimated UMLR model displaying a good fit, an acceptable fit or a poor fit to the observations? Obviously the answer to such a question is either much too complicated or much too simple. First of all, the answer depends strongly on the application at hand and the purpose of the regression analysis. Next, the answer is “hidden” in many of the (interrelated) parameters and quantities discussed in previous sections. These include

- the coefficient of multiple determination, R^2
- the adjusted coefficient of multiple determination, R_{adj}^2
- the number of observations, n
- the p -value for the test of an overall significant regression model (ANOVA-table)
- the various properties of the estimated residuals, $\hat{e}_i$
- and others ...

Thus, to make a simple decision rule for detecting a good, acceptable or poor fit, will, by definition, be somewhat arbitrary and without solid scientific foundation. Nevertheless, an attempt will be made in the following, first for the overall fit, and secondly for the importance of the individual predictor variables (section 14).

The two parameters selected for developing a simple, heuristic decision rule will be the adjusted coefficient of multiple determination, R_{adj}^2 , and the number of observations, n . As stated in section 8, R_{adj}^2 is a more robust performance parameter than R^2 , since it to some degree incorporates the number of predictor variables, k , and also the number of observations, n . To some degree here means that R_{adj}^2 as opposed to R^2 is an unbiased estimator.

However, the interpretation of the value of R_{adj}^2 should depend on n representing either a highly sufficient sample of observations, a moderately sufficient sample or a very small sample.

It might correctly be stated, that the p -value for the test of an overall significant regression model (ANOVA-table) in a way contains the information from both R_{adj}^2 and n . However, since p -values are numerically much more spread (typically many decades), it is a little harder to “discretize” in a reasonable way.

Thus, here it is selected to use the combination of R_{adj}^2 and n .

When, then, is a sample “highly sufficient” in size? Again, of course, no unique answer can be given. However, a value $n \geq 30$ is often used as a rule of thumb. Suggestively, a “good fit” for a highly sufficient sample could then be defined to be, e.g. $R_{adj}^2 \geq 0.75$, since this corresponds to a coefficient of multiple correlation $r \geq \text{approx. } 0.85$. An “acceptable fit” could be defined to be $0.65 \leq R_{adj}^2 < 0.75$, since it corresponds to a coefficient of multiple correlation $\text{approx. } 0.80 \leq r < \text{approx. } 0.85$. Finally a “poor fit” could then be defined to be $R_{adj}^2 < 0.65$, since this corresponds to a coefficient of multiple correlation $r < \text{approx. } 0.80$.

A “moderately sufficient” sample will here then be defined to be the interval $15 \leq n < 30$. Perhaps the lower limit here is a little too liberal, but will be used anyway. Suggestively, a “good fit” for the minimal moderately sufficient sample, i.e. for $n = 15$, could then be defined to be, e.g. $R_{adj}^2 \geq 0.80$, an “acceptable fit” for this case could be defined to be $0.70 \leq R_{adj}^2 < 0.80$, and a “poor fit” could be defined to be $R_{adj}^2 < 0.70$.

Finally, a “very small” sample will, again perhaps being a little to the liberal side, be defined to be the interval $5 \leq n < 15$. Suggestively, a “good fit” for the absolutely minimal sample, i.e. for $n = 5$, could then be defined to be, e.g. $R_{adj}^2 \geq 0.90$, an “acceptable fit” for this case could be defined to be $0.80 \leq R_{adj}^2 < 0.90$, and a “poor fit” could be defined to be $R_{adj}^2 < 0.80$.

For $n < 5$, a regression analysis will be more or less meaningless, and will be considered as such here.

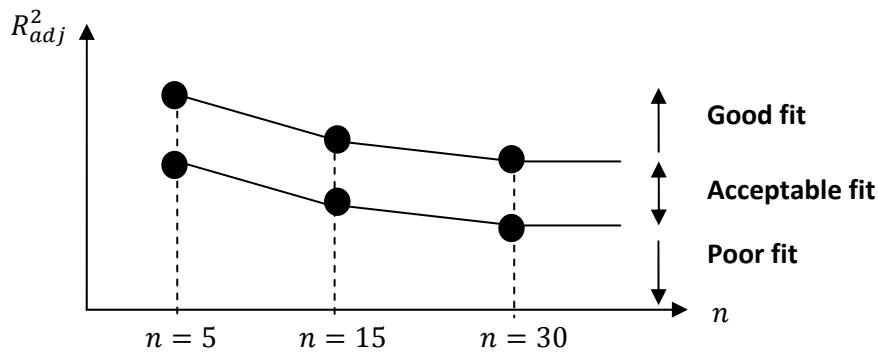
Next, to interpolate between the sample sizes $n = 30$, $n = 15$ and $n = 5$, respectively, of course many schemes could be suggested. Here, however, lacking arguments for making it more complicated, is suggested to use a simple linear interpolation scheme.

These considerations can be summarized in the table below.

Heuristic decision rule for overall fit of regression model:

	Good fit	Acceptable fit	Poor fit
$n \geq 30$	$R_{adj}^2 \geq 0.75$	$0.65 \leq R_{adj}^2 < 0.75$	$R_{adj}^2 < 0.65$
$15 \leq n < 30$	$R_{adj}^2 \geq 0.75 + 0.05 \frac{30-n}{15}$	$R_{adj}^2 < 0.75 + 0.05 \frac{30-n}{15}$ and $R_{adj}^2 \geq 0.65 + 0.05 \frac{30-n}{15}$	$R_{adj}^2 < 0.65 + 0.05 \frac{30-n}{15}$
$5 \leq n < 15$	$R_{adj}^2 \geq 0.80 + 0.10 \frac{15-n}{10}$	$R_{adj}^2 < 0.80 + 0.10 \frac{15-n}{10}$ and $R_{adj}^2 \geq 0.70 + 0.10 \frac{15-n}{10}$	$R_{adj}^2 < 0.70 + 0.10 \frac{15-n}{10}$

Alternatively, the suggested heuristic decision rule can be visualized as in the figure below



Of course, there is no really good argument for the change of slope for the shown curves at $n = 15$ and $n = 30$, but it is a consequence of the above heuristic considerations.

14 Marginal heuristic rating of predictor variables

To evaluate the importance of the individual predictor variables, an intuitive approach would be to decompose the previously defined Sum of Squares of Regression, SSR , into components corresponding to each of the k predictor variables. Unfortunately, in general, this is not possible due to interaction terms between predictor variables in the sum. A more or less equivalent approach to this will be described in Part E2.

Here, however, a simple heuristic scheme for evaluating the individual predictor variables will be suggested, using only the regression table's marginal tests described in section 10. Each predictor variable will be classified as being either "very important", "moderately important" or "unimportant". The obvious candidate parameter for this classification is the p -value for each of the marginal tests for significant predictor variables.

Heuristic classification for individual predictor variables:

For each predictor variable x_j , $j = 0, 1, \dots, k$:

Calculate the p -value for the corresponding marginal test for significance:

$$p_j = 2P(t_{n-k-1} > |t_{test, \beta_j}|) \quad \text{where the test statistic is } t_{test, \beta_j} = \hat{\beta}_j / s_{\hat{\beta}_j}$$

If $p_j \leq 0.05$ then predictor variable x_j is very important

If $0.05 < p_j \leq 0.10$ then predictor variable x_j is moderately important

If $p_j > 0.10$ then predictor variable x_j is unimportant

Of course, the indicated p -value limits are somewhat arbitrary and can be modified as desired in an actual application context.

Note that the available number of observations, n , is incorporated in the p -value calculations.

Finally, this classification scheme is only making sense, if the overall fit of the regression model, as described in the previous section, is classified as being a "good fit" or at least an "acceptable fit".

15 Regression through origin and related problems

The general UMLR model considered so far

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + e$$

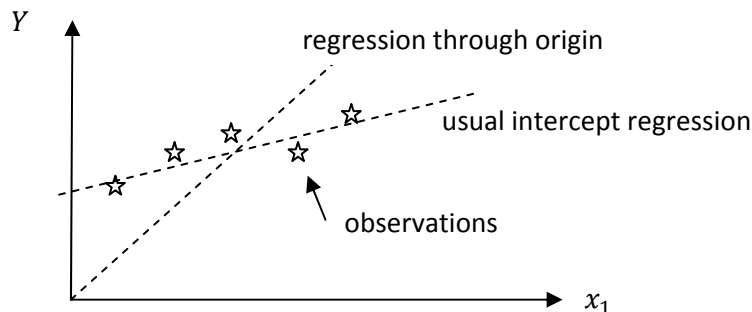
includes an intercept (or “constant”) term β_0 .

Alternatively the model can be defined without the intercept term and is then usually referred to as “regression through origin”

$$Y = \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + e$$

This is a completely different model from the one studied so far, and all results concerning estimation of parameters, model control, hypothesis testing on regression coefficients and calculation of confidence intervals will change compared to the usual intercept regression model, even though the interpretation of the results are unchanged.

As an example of this, for simple linear regression it is easy to visualize, as shown below, that the Least Squares estimated slope regression coefficient, β_1 , is not the same for regression through origin and intercept regression, given the same observations. This generalizes directly to the multiple predictor variables case.



Besides the need for a new set of results for regression through origin, there are several intrinsic problems with this model, including the following

- If an intercept term is ignored, the remaining regression coefficients will be poorly estimated, and therefore any statistical statement about them will be misleading
- The estimated residuals will generally not sum to zero, $\sum_i \hat{e}_i \neq 0$
- The coefficient of multiple determination, R^2 , which is often an important part of evaluating the fit of the regression model, can for regression through origin in fact be negative, and in general its interpretation is not straightforward, unless the usual Sums of Squares of Regression and of Error, SSR and SSE respectively, are redefined to be “uncorrected”, i.e. without the usual form subtraction of means.

For these reasons and also other complications, it is good practice, only to use the regression through origin model, if there is clear scientifically based evidence for it. An imagined example of such a situation could be when observing (measuring) the gravitational force on a number of solid bodies with equal spatial topology and surface, but with different masses, since in this situation the scientifically based (approximate first-order) regression model is $F_i = g \cdot m_i + e_i$.

Thus in general, when fitting a regression model to observations, it is recommendable always first to use the general intercept regression model, and only if the estimated intercept term is numerically much less than response variable observations, to consider a regression through origin model.

PART E1: Extensions 1:

Simultaneous inference for predictor variables

16 Simultaneous inference using approximations

In section 10 the approach for making inferences on regression coefficients – based on the regression table - was as stated to do it in a univariate setting, which is equivalent to do marginal inferences on the coefficients, i.e. to address them one-at-a-time while disregarding the others. On the other hand, if it is desired to consider and make statistical statements – such as hypothesis tests or confidence intervals - about several or all of the coefficients simultaneously, the marginal inference approach must be modified. This will be discussed in this and the following section. In this section the approach will be to use a popular approximation, called the Bonferroni correction, while in the next section the approach will be genuinely multivariate.

The problem with making simultaneous statistical statements about several or all regression coefficients based on the regression table's marginal hypothesis tests and confidence intervals is, that the specified level of significance, α , or equivalently the specified level of confidence $(1 - \alpha)100\%$, is not simultaneously achieved for the set of considered coefficients. Thus the actual value of α is not equal to the desired or specified value.

Remember, that specifying a particular value of α is equivalent to specifying the acceptable probability of a type I error, that is, an acceptable probability of rejecting the null-hypothesis, H_0 , when in fact H_0 is true

$$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \mid H_0 \text{ true})$$

However, it turns out that when making simultaneous tests on several or all coefficients, the effectively achieved type I error probability, call it α_{eff} , is not equal to α , in fact $\alpha_{\text{eff}} > \alpha$.

To see this, assume for a moment that all $k + 1$ regression coefficients $\beta_0, \beta_1, \dots, \beta_k$ are considered simultaneously. The collective, or simultaneous, type I error probability is then the probability that at least one of the null-hypotheses

$$H_{0,j}: \beta_j = 0, \quad j = 0, 1, \dots, k$$

is mistakenly rejected when, in fact, they were all true, that is

$$\begin{aligned} \alpha_{\text{eff}} &= P(\text{simultaneous type I error}) = P(\text{reject at least one } H_{0,j} \mid \text{all } H_{0,j} \text{ true}) \\ &= 1 - P(\text{accept all } H_{0,j} \mid \text{all } H_{0,j} \text{ true}) \end{aligned}$$

Next, assume that the $k + 1$ estimated regression coefficients $\hat{\beta}_0 \hat{\beta}_1 \dots \hat{\beta}_k$ are independent and thereby also uncorrelated. Then the calculation proceeds as

$$\begin{aligned}
 \alpha_{\text{eff}} &= P(\text{simultaneous type I error}) = 1 - \prod_{j=0}^k P(\text{accept } H_{0,j} \mid H_{0,j} \text{ true}) \\
 &= 1 - \prod_{j=0}^k (1 - P(\text{reject } H_{0,j} \mid H_{0,j} \text{ true})) \\
 &= 1 - \prod_{j=0}^k (1 - \alpha) \\
 &= 1 - (1 - \alpha)^{k+1} \\
 &> \alpha
 \end{aligned}$$

The conclusion is that the effective type I error rate, sometimes referred to as the *experiment-wise error rate*, is higher than specified. This means for simultaneous hypothesis testing, that the null-hypotheses are rejected too often, and for simultaneous confidence intervals, that the level of confidence is too low, or equivalently, the confidence intervals are too narrow.

An easy, approximate way to address this problem is to use the so called Bonferroni correction. To introduce this, note that, assuming as typically $\alpha \ll 1$,

$$\alpha_{\text{eff}} = P(\text{simultaneous type I error}) = 1 - (1 - \alpha)^{k+1} \cong (k + 1) \alpha$$

Thus, given that a desired value of α is specified, if a modified value $\alpha_{\text{mod}} = \frac{\alpha}{k+1}$ is used instead, then

$$\alpha_{\text{eff}} = P(\text{simultaneous type I error}) \cong (k + 1) \alpha_{\text{mod}} = (k + 1) \frac{\alpha}{k+1} = \alpha$$

that is, the specified level of significance, and thereby the specified type I error probability is approximately simultaneously achieved for all $k + 1$ regression coefficients $\beta_0, \beta_1, \dots, \beta_k$.

To use the Bonferroni correction for regression coefficients confidence intervals, simply substitute α with the value $\alpha_{\text{mod}} = \frac{\alpha}{k+1}$ in the interval calculations giving

$$\begin{aligned}
 &(1 - \alpha)100\% \text{ Bonferroni confidence intervals for } \beta_j, j = 0, 1, \dots, k : \\
 &\left[CI_{\beta_j, \text{low}, \alpha}, CI_{\beta_j, \text{high}, \alpha} \right] = \left[\hat{\beta}_j \pm s_{\hat{\beta}_j} \cdot t_{n-(k+1), \frac{\alpha}{2(k+1)}} \right]
 \end{aligned}$$

The Bonferroni confidence intervals will always be wider than the marginal confidence intervals because of the increased quantile multiplicand

$$t_{n-(k+1), \frac{\alpha}{2(k+1)}} > t_{n-(k+1), \frac{\alpha}{2}}$$

Likewise, to use the Bonferroni correction for simultaneous test of the null-hypotheses

$$H_{0,j}: \beta_j = 0, j = 0, 1, \dots, k$$

simply substitute α with the value $\alpha_{\text{mod}} = \frac{\alpha}{k+1}$ in the accept/reject decision rules giving the procedure

- Choose level of significance as the (small) fraction α , or equivalently $100 \cdot \alpha \%$
- For $j = 0, 1, \dots, k$:
- Calculate the test statistic value $t_{\text{test}, \beta_j} = \frac{\hat{\beta}_j}{s_{\hat{\beta}_j}}$ based on the observations data
- If $|t_{\text{test}, \beta_j}| > t_{n-(k+1), \frac{\alpha}{2(k+1)}}$ reject $H_{0,j}$
- If $|t_{\text{test}, \beta_j}| \leq t_{n-(k+1), \frac{\alpha}{2(k+1)}}$ accept $H_{0,j}$

The Bonferroni approximation can be used in the same way as above if only a subset, but not all, of the regression coefficients are considered simultaneously. For example, if g coefficients are considered, where $1 < g < k + 1$, then the factor $k + 1$ in the above equations are simply substituted by g .

17 True multivariate simultaneous inference

The problem with the Bonferroni approximation is that it assumes the $k + 1$ regression coefficient estimates $\hat{\beta}_0 \hat{\beta}_1 \dots \hat{\beta}_k$ to be independent. However, the vector of estimated coefficients has as stated previously a joint, multivariate Gaussian distribution

$$\hat{\boldsymbol{\beta}} = [\hat{\beta}_0 \hat{\beta}_1 \dots \hat{\beta}_k]^T \sim MVN(\boldsymbol{\beta}, \boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}})$$

with variance-covariance matrix

$$\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}} = \begin{bmatrix} V[\hat{\beta}_0] & Cov[\hat{\beta}_0, \hat{\beta}_1] & \dots & Cov[\hat{\beta}_0, \hat{\beta}_k] \\ Cov[\hat{\beta}_0, \hat{\beta}_1] & V[\hat{\beta}_1] & & \vdots \\ \vdots & \vdots & & \vdots \\ Cov[\hat{\beta}_0, \hat{\beta}_k] & \dots & \dots & V[\hat{\beta}_k] \end{bmatrix} = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

and, in general, the estimated regression coefficients are not uncorrelated and therefore not independent. The Bonferroni approximation is equivalent to assuming $\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}$ to be diagonal, which, in general, it is not. Thus, taking this into account, the inferences on regression coefficients should be genuinely multivariate even though the regression model is univariate in the response variable Y .

The multivariate, simultaneous inferences on regression coefficients are based on the following distribution result

$$\frac{(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T \mathbf{S}_{\hat{\boldsymbol{\beta}}}^{-1} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{k + 1} \sim F_{k+1, n-(k+1)}$$

where $\mathbf{S}_{\hat{\boldsymbol{\beta}}}$ is the estimated variance-covariance matrix for $\hat{\boldsymbol{\beta}}$

$$\mathbf{S}_{\hat{\boldsymbol{\beta}}} = \hat{\boldsymbol{\Sigma}}_{\hat{\boldsymbol{\beta}}} = \begin{bmatrix} \hat{V}[\hat{\beta}_0] & \widehat{Cov}[\hat{\beta}_0, \hat{\beta}_1] & \dots & \widehat{Cov}[\hat{\beta}_0, \hat{\beta}_k] \\ \widehat{Cov}[\hat{\beta}_0, \hat{\beta}_1] & \hat{V}[\hat{\beta}_1] & & \vdots \\ \vdots & \vdots & & \vdots \\ \widehat{Cov}[\hat{\beta}_0, \hat{\beta}_k] & \dots & \dots & \hat{V}[\hat{\beta}_k] \end{bmatrix} = \hat{\sigma}^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

This result can be used to define first a confidence region for the vector of unknown regression coefficients, $\boldsymbol{\beta}$, and based on this to define simultaneous confidence intervals for the individual coefficients.

The confidence region for $\boldsymbol{\beta}$ is a hyper ellipsoid in the $(k + 1)$ -dimensional space where the individual coefficients $\beta_0, \beta_1, \dots, \beta_k$ are the axes.

Based on the above stated distribution result can be seen

$$P \left(\frac{(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T \mathbf{S}_{\hat{\boldsymbol{\beta}}}^{-1} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{k + 1} \leq F_{k+1, n-(k+1), \alpha} \right) = 1 - \alpha$$

This means that

$(1 - \alpha)100\%$ simultaneous confidence region for $\boldsymbol{\beta}$ is all points $\boldsymbol{\beta} = [\beta_0 \ \beta_1 \ \dots \ \beta_k]^T$ in $(k + 1)$ -dimensional space within the hyper ellipsoid

$$\frac{(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T \mathbf{S}_{\hat{\boldsymbol{\beta}}}^{-1} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{k + 1} = F_{k+1, n-(k+1), \alpha}$$

The hyper ellipsoid is centered at $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ and for simple linear regression it simplifies to a usual plane ellipsis.

The multidimensionality of the confidence region makes it of limited practical use. Therefore can instead be stated simultaneous confidence intervals for the individual regression coefficients. These intervals are the projections (or the “shadows”) of the hyper ellipsoid on the axes and can be calculated as

$$(1 - \alpha)100\% \text{ simultaneous confidence intervals for } \beta_j, j = 0, 1, \dots, k : \\ [CI_{\beta_j, low, \alpha}, CI_{\beta_j, high, \alpha}] = \left[\hat{\beta}_j \pm s_{\hat{\beta}_j} \cdot \sqrt{(k + 1) F_{k+1, n-(k+1), \alpha}} \right]$$

The simultaneous confidence intervals will always be wider (more conservative) than the Bonferroni confidence intervals, which in turn, as stated in the previous section, will always be wider than the marginal confidence intervals. This is because of the following size order of the involved quantile multiplicands

$$\sqrt{(k+1) F_{k+1, n-(k+1), \alpha}} > t_{n-(k+1), \frac{\alpha}{2(k+1)}} > t_{n-(k+1), \frac{\alpha}{2}}$$

In practice, the Bonferroni confidence intervals are perhaps the most often used form of simultaneous confidence intervals, since they are easy to find and is a reasonable compromise between being too liberal or too conservative.

The stated distribution result can also be used to devise a simultaneous, multivariate hypothesis test for the vector of regression coefficients β . To test if β might attain a particular hypothesized value, say μ_0 , that is to test

$$\begin{aligned} H_0: \beta &= \mu_0 \\ H_1: \beta &\neq \mu_0 \end{aligned}$$

can be used the test statistic

$$F_{test} = \frac{(\hat{\beta} - \mu_0)^T S_{\hat{\beta}}^{-1} (\hat{\beta} - \mu_0)}{k+1} \stackrel{H_0}{\approx} F_{k+1, n-(k+1)}$$

In particular, to test if the vector of regression coefficients is significantly different from the $\mathbf{0}$ -vector

$$\begin{aligned} H_0: \beta &= \mathbf{0} \\ H_1: \beta &\neq \mathbf{0} \end{aligned}$$

use the test statistic

$$F_{test} = \frac{\hat{\beta}^T S_{\hat{\beta}}^{-1} \hat{\beta}}{k+1} \stackrel{H_0}{\approx} F_{k+1, n-(k+1)}$$

The hypothesis test procedure is then

- Choose level of significance as the (small) fraction α , or equivalently $100 \cdot \alpha \%$
- Calculate the test statistic value $F_{test} = \frac{\hat{\beta}^T S_{\hat{\beta}}^{-1} \hat{\beta}}{k+1}$ based on the observations data
- If $F_{test} > F_{k+1, n-(k+1), \alpha}$ reject H_0
- If $F_{test} \leq F_{k+1, n-(k+1), \alpha}$ accept H_0

Tests can also be devised for subsets of the regression coefficients instead of the entire vector, e.g. to test if a specific subset of the coefficients is significantly different from zero. This involves so called likelihood ratio tests, a subject which will not be addressed further here.

PART E2: Extensions 2:

Search for optimal subset of predictor variables

18 Improved model simplification procedure using the Extra Sums of Squares concept

The simple procedure for screening of predictor variables, model simplification and optimization proposed in section 11 is based on the p -values for the marginal tests on significant predictor variables, i.e. the tests from the regression table.

This procedure is not optimal in several ways.

First, as stated in section 11, it does not guarantee that the “best” of all possible regression models for the pre-defined set of predictor variables is found.

Secondly, since it is based on marginal tests, it ignores the points just addressed in sections 16 and 17, i.e. the problems stemming from using a one-variable-at-a-time approach for an inherently simultaneous-variables situation.

Finally, it is based on hypothesis test p -values and not on the more robust decomposition of the total sum of squares of observation variations.

In this section is discussed an alternative, improved procedure, which will remove the third of these disadvantages of the procedure of section 11, and also, to some degree, remove the second. The price to be paid for this is an increased complexity in the procedure itself, as well as in the required numerical computations.

Afterwards, In the next section, will be discussed an exhaustive search procedure, which according to some criteria finds the “best” regression model, thus also removing the first of the mentioned disadvantages.

Recall that the total response variable variation sum of squares can be decomposed as

$$\begin{aligned} SST &= \sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \\ &= SSE + SSR \end{aligned}$$

The total sum of squares, SST , is constant for a given set of response variable observations, no matter how many predictor variables are used.

The sum of squares of regression, SSR , however, will vary in value depending on how many predictor variables are involved in the calculation of the response variable estimate, $\hat{Y}_i$, and since SST is constant, the sum of squares of error, SSE , will also vary.

Therefore, since in this section the number of predictor variables will be varying, to make the discussion unambiguous, the notation $SSR(x_1, \dots, x_k)$, and likewise $SSE(x_1, \dots, x_k)$, will be used, where the set of variables in the parentheses indicates which predictor variables are presently used in the regression model.

The sum of squares of regression, $SSR(x_1, \dots, x_k)$, is the part of SST , that explains the response variable's variation by the influence of the predictor variables $x_1, \dots, x_k$.

The first of two problems with directly using the sum of squares decomposition for model simplification, is that even though it would have been very nice, $SSR(x_1, \dots, x_k)$ can not directly be decomposed into k parts describing the influence of each individual predictor variable, since

$$\begin{aligned} SSR(x_1, \dots, x_k) &= \sum_{i=1}^n ([\hat{\beta}_0 + \hat{\beta}_1 x_{i1} \dots + \hat{\beta}_k x_{ik}] - \bar{Y})^2 \\ &\neq SSR(x_1) + \dots + SSR(x_k) \\ &= \sum_{i=1}^n ([\hat{\beta}_0 + \hat{\beta}_1 x_{i1}] - \bar{Y})^2 + \dots + \sum_{i=1}^n ([\hat{\beta}_0 + \hat{\beta}_k x_{ik}] - \bar{Y})^2 \end{aligned}$$

The reason for the un-equality sign is that in the first of the above sums, interaction terms between predictor variables will contribute to the total sum (unless the predictor variable values can a priori be designed to be *orthogonal*, which is usually not the case).

The second problem with directly using the sum of squares decomposition for considering various, different sets of predictor variables, is that when increasing the number of predictor variables, SSE can not increase (i.e. will remain unchanged or typically decrease), since better estimates of the response variable observations are generally achieved. Again, since SST is constant, SSR can not decrease (i.e. will remain unchanged or typically increase) when increasing the number of predictor variables.

However, are these increases in SSR big enough to justify adding the extra predictor variables to the model, thus making it more complex and also loosing degrees of freedom for error variance estimation ?

Both of these sum of squares decomposition problems is addressed by the auxiliary concept Extra Sums of Squares, which will now be introduced and afterwards used.

First, for a definition of an Extra Sum of Squares, assume that initially only the predictor variable x_1 is used in the regression model, giving

$$SSR(x_1) = \sum_{i=1}^n ([\hat{\beta}_{0,x_1} + \hat{\beta}_{1,x_1} x_{i1}] - \bar{Y})^2$$

Then a second predictor variable x_2 is included in the model, giving

$$SSR(x_1, x_2) = \sum_{i=1}^n ([\hat{\beta}_{0,x_1,x_2} + \hat{\beta}_{1,x_1,x_2}x_{i1} + \hat{\beta}_{2,x_1,x_2}x_{i2}] - \bar{Y})^2 \geq SSR(x_1)$$

The extra sum of squares is then defined as

$$SSR(x_2|x_1) \stackrel{\text{def}}{=} SSR(x_1, x_2) - SSR(x_1)$$

where the vertical bar in the definition should be read as “given” or “conditioned on” the hereafter listed predictor variable(s) already being in the model.

Alternatively put, $SSR(x_1, x_2)$ for both predictor variables in the model can be decomposed as

$$SSR(x_1, x_2) = SSR(x_1) + SSR(x_2|x_1)$$

Of course the order of introduction of predictor variables in the model could be permuted giving the alternative decomposition

$$SSR(x_1, x_2) = SSR(x_2) + SSR(x_1|x_2)$$

If the two considered predictor variables x_1, x_2 are highly correlated, the increase in SSR , when introducing the second variable given that the first is already included, will be relatively small

$$SSR(x_2|x_1) \ll SSR(x_1) \quad \text{and} \quad SSR(x_1|x_2) \ll SSR(x_2)$$

Equivalently, the influence of a particular predictor variable on the value of SSR will be much smaller when a correlated predictor variable is already in the model, compared to if the particular predictor variable was alone in the model

$$SSR(x_1|x_2) \ll SSR(x_1) \quad \text{and} \quad SSR(x_2|x_1) \ll SSR(x_2)$$

The extra sum of squares definition generalizes directly to three or more predictor variables.
Thus, introducing a third predictor variable x_3 in the model yields

$$\begin{aligned} SSR(x_1, x_2, x_3) &= \sum_{i=1}^n ([\hat{\beta}_{0,x_1,x_2,x_3} + \hat{\beta}_{1,x_1,x_2,x_3}x_{i1} + \hat{\beta}_{2,x_1,x_2,x_3}x_{i2} + \hat{\beta}_{3,x_1,x_2,x_3}x_{i3}] - \bar{Y})^2 \\ &\geq SSR(x_1, x_2) \end{aligned}$$

and the extra sum of square, given that x_1, x_2 are already in the model, is then defined as

$$SSR(x_3|x_1, x_2) \stackrel{\text{def}}{=} SSR(x_1, x_2, x_3) - SSR(x_1, x_2)$$

The extra sum of square concept can be used for performing hypothesis tests on the significance of a single, several or all predictor variables, and since this will be a part of the improved model simplification procedure to be proposed shortly, such tests will be discussed in brief below.

Assuming as usual that all error terms are Gaussian distributed, all sums of squares, including the extra sums of squares, will be chi-square distributed. Dividing the sums of squares with their respective degrees of freedom yields variance estimates. As used in previous sections these variance estimates are also called mean squares, where estimates for regression variance is termed MSR and estimates for error variance is termed MSE . All test-statistics below can be seen as the ratio between an MSR and an MSE , and they will all, under the stated assumptions on error terms, have F -distributions with degrees of freedom according to the MSR numerator and the MSE denominator.

First then, to test the significance of all predictor variables together, or equivalently to test the hypothesis on all regression coefficients together (except the intercept term)

$$\begin{aligned} H_0: & \beta_1 = \beta_2 = \dots \beta_k = 0 \\ H_1: & \exists j \in \{1, 2, \dots, k\}: \beta_j \neq 0 \end{aligned}$$

is calculated the test-statistic

$$F_{test} = \frac{MSR(x_1, \dots, x_k)}{MSE(x_1, \dots, x_k)} = \frac{SSR(x_1, \dots, x_k)/k}{SSE(x_1, \dots, x_k)/(n - k - 1)} \sim F_{k, n-k-1}$$

where

$$SSE(x_1, \dots, x_k) \stackrel{\text{def}}{=} SST - SSR(x_1, \dots, x_k)$$

This is exactly the test that was introduced in section 8 and called test for overall significance of the regression model, and it is only repeated here for completeness. As always for tests in F -distributions, a high value of the test-statistic indicates that it is unlikely that the null-hypotheses should in fact be true.

More precisely, using the p -value approach to hypothesis testing explained previously,

$$p\text{-value} = P(F_{k, n-k-1} > F_{test})$$

Next, to test the significance of single particular predictor variable, or equivalently to test a hypothesis on a particular regression coefficient (here β_1 is used as example)

$$\begin{aligned} H_0: & \beta_1 = 0 \\ H_1: & \beta_1 \neq 0 \end{aligned}$$

is calculated a test-statistic, where, as can be seen, the numerator is in fact based on the extra sum of squares for introducing predictor variable x_1 into the model, given that all the other predictor variables $x_2, \dots, x_k$ are already in the model

$$F_{test} = \frac{MSR(x_1|x_2, \dots, x_k)}{MSE(x_1, \dots, x_k)} = \frac{SSR(x_1|x_2, \dots, x_k)/1}{SSE(x_1, \dots, x_k)/(n - k - 1)} \sim F_{1, n-k-1}$$

The conclusion of this test can then for example be based on the p -value

$$p\text{-value} = P(F_{1, n-k-1} > F_{test})$$

This type of test on a single predictor variable is mathematically equivalent to the regression table's marginal tests introduced in section 10, even though in that section the test-statistics were t -distributed, while in the present setting they are F -distributed.

Again, using the extra sums of squares concept, tests can be devised for the simultaneous significance of a number of predictor variables. For example, to perform the simultaneous test

$$\begin{aligned} H_0: \beta_1 &= \beta_2 = 0 \\ H_1: \beta_1 &\neq 0 \vee \beta_2 \neq 0 \end{aligned}$$

is calculated the extra sum of squares based test-statistic

$$F_{test} = \frac{MSR(x_1, x_2|x_3, \dots, x_k)}{MSE(x_1, \dots, x_k)} = \frac{SSR(x_1, x_2|x_3, \dots, x_k)/2}{SSE(x_1, \dots, x_k)/(n - k - 1)} \sim F_{2, n-k-1}$$

and the p -value

$$p\text{-value} = P(F_{2, n-k-1} > F_{test})$$

It should be evident how this methodology using extra sums of squares generalizes easily to any subset of the predictor variables.

Note also that the F -tests involving more than a single predictor variable have no t -test counterpart.

At last, all is ready for stating the proposed improved model simplification procedure based on the extra sum of squares concept and the type of hypothesis tests just described.

For ease of interpretation, the procedure will be explained using an example, where it is assumed that the four predictor variables x_1, x_2, x_3, x_4 are considered candidates for the regression model based on n observations. The example generalizes in an obvious way to any number of predictor variable candidates.

The procedure is based on a principle which may be termed “stepwise regression using forward selection of predictor variables”. Thus, as will be evident, the procedure is iterative.

Step 1: introduce first variable into model and validate

Calculate the extra sums of squares $SSR(x_1), SSR(x_2), SSR(x_3), SSR(x_4)$

Select the largest, here it is assumed, that it is $SSR(x_1)$

Thus predictor variable x_1 is introduced into the model

Test the significance of adding x_1 to the model using the test-statistic

$$F_{test} = \frac{SSR(x_1)/1}{SSE(x_1)/(n-2)} \sim F_{1,n-2}$$

Use either p -value or level of significance approach for test conclusion

If the test is significant: include x_1 in model

Otherwise the procedure is finished, no predictor variables are significant

Step 2: introduce second variable into model and validate

Calculate the extra sums of squares $SSR(x_2|x_1), SSR(x_3|x_1), SSR(x_4|x_1)$

Select the largest, here it is assumed, that it is $SSR(x_2|x_1)$

Thus predictor variable x_2 is introduced into the model

Test the significance of adding x_2 to the model, given that x_1 is already in the model, using the test-statistic

$$F_{test} = \frac{SSR(x_2|x_1)/1}{SSE(x_1, x_2)/(n-3)} \sim F_{1,n-3}$$

Use either p -value or level of significance approach for test conclusion

If the test is significant: include x_2 in model along with x_1

Otherwise the procedure is finished, only x_1 is significant in model

Step 3: introduce third variable into model and validate

Calculate the extra sums of squares $SSR(x_3|x_1, x_2), SSR(x_4|x_1, x_2)$

Select the largest, here it is assumed, that it is $SSR(x_3|x_1, x_2)$

Thus predictor variable x_3 is introduced into the model

Test the significance of adding x_3 to the model, given that x_1, x_2 are already in the model, using the test-statistic

$$F_{test} = \frac{SSR(x_3|x_1, x_2)/1}{SSE(x_1, x_2, x_3)/(n-4)} \sim F_{1,n-4}$$

Use either p -value or level of significance approach for test conclusion

If the test is significant: include x_3 in model along with x_1, x_2

Otherwise the procedure is finished, only x_1, x_2 are significant in model

Step 4,5,... : introduce fourth, fifth, ... variable into model and validate

In this way the iterative procedure continues until either the most recent predictor variable candidate fails to be significant or all predictor variable candidates are included in the model.

Further improvement of procedure

The stepwise regression procedure can optionally be further improved by complementing the forward selection of predictor variables principle with a backward elimination principle, which can be seen as a kind of cross validation of variables.

This means that every time a new predictor variable is found significant in one of the steps described above, and thus included in the model, all previously included predictor variables in the model are validated for still being significant, even with the new variable also being present.

For example if, from step 1 above, x_1 is already included in the model, and in step 2 x_2 is also found significant and thus included, a supplementary test on the “reverse” extra sum of square $SSR(x_1|x_2)$ determines if x_1 still should be kept in the model or removed.

This extra cross validation of predictor variables decreases the probability that highly correlated predictor variables are allowed in the model simultaneously, thereby making the model more complex than necessary. But, of course, the supplementary backward elimination tests also increases the complexity of the procedure.

19 Exhaustive search for optimal subset

Given an UMLR model with k pre-selected predictor variable candidates $x_1, x_2, \dots, x_k$

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k,$$

the model simplification procedures of section 11 and 18 attempt in two different ways to reach at a “good” model with a subset (or all) of the candidate variables included. Neither of the procedures guarantees, however, to find the “best” model with respect to some easy definable criterium.

If it is indeed desired to find a “best” model, an exhaustive search of all possible subsets of predictor variables must be performed, comparing the performance of all these with respect to a pre-defined optimization criterium.

With k predictor variable candidates, the total number of subsets including $0, 1, 2, \dots, k$ of the variables is

$$\binom{k}{0} + \binom{k}{1} + \dots + \binom{k}{k} = 2^k$$

and thus, growing exponentially with k . Therefore an exhaustive search is only realistic for moderate values of k , say e.g. for $k \leq 5 - 6$.

Several optimization criteria are suggested in the literature, the simplest being the previously defined adjusted coefficient of multiple determination

$$R_{adj}^2 = 1 - \left(\frac{n-1}{n-k-1} \right) \frac{SSE}{SST}, \quad k = \text{number of variables in subset}$$

Thus, for each of the possible subsets of predictor variables, the ANOVA table is generated, R_{adj}^2 is calculated, and the subset with highest value of R_{adj}^2 is selected for the model.

PART E3: Extensions 3: UMLR model generalizations

20 Weighted Least Squares model

When analyzing the estimated residuals as described in section 6, one of the often encountered violations of the assumptions of the basic UMLR model is that the variance of the error terms is not constant as assumed, but varies with the value of response and predictor variable values.

Recall that the basic model is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + e_i = \mathbf{x}_i \boldsymbol{\beta} + e_i, \quad i = 1, 2, \dots, n$$

Since the error terms are assumed to have constant variance, the variance of the response variable is also assumed to be constant

$$V[e_i] = \sigma^2 \Rightarrow V[Y_i] = \sigma^2$$

Combining this assumption with the assumption of independent error terms, the variance-covariance matrix of the error terms and of the response variable is thus assumed to be proportional to the identity matrix

$$\boldsymbol{\Sigma}_e = \boldsymbol{\Sigma}_y = \begin{bmatrix} \sigma^2 & 0 & \dots \\ 0 & \ddots & 0 \\ \vdots & 0 & \sigma^2 \end{bmatrix} = \sigma^2 \mathbf{I}$$

However, as indicated above, often the variance is not constant but varies with the observation index

$$V[e_i] = \sigma_i^2 \Rightarrow V[Y_i] = \sigma_i^2$$

If this tendency is very marked and there are very few observations, e.g. $n < 20 - 30$, a so called Weighted Least Squares approach must be used. This requires some modifications of the basic model as discussed in this section.

Allowing varying variances, the modified model can be stated

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + e_i = \mathbf{x}_i \boldsymbol{\beta} + e_i, \quad i = 1, 2, \dots, n$$

$$E[e_i] = 0 \quad V[e_i] = V[Y_i] = \sigma_i^2$$

Assuming independent error terms as before, the modified variance-covariance matrices will still be diagonal, but now with possibly different diagonal values

$$\boldsymbol{\Sigma}_e = \boldsymbol{\Sigma}_y = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2) = \begin{bmatrix} \sigma_1^2 & 0 & \dots \\ 0 & \ddots & 0 \\ \vdots & 0 & \sigma_n^2 \end{bmatrix}$$

In this situation, the Ordinary Least Squares approach is not optimal. Instead, the Weighted Least Squares approach works by weighting each element of the sum of squares - minimized in the least squares procedure - according to its variance. Thus, high weights should be given to elements with low variance (low expected deviation from the unknown, ideal regression equation), while low weights should be given to elements with high variance (high expected deviation from the unknown, ideal regression equation), as intuitively correct.

More precisely, each observation should be weighted with the reciprocal of its variance

$$w_i = \frac{1}{V[Y_i]} = \frac{1}{\sigma_i^2}, \quad i = 1, 2, \dots, n$$

The weighted sum of squares of estimated residuals is then

$$SSE = \sum_{i=1}^n \frac{(Y_i - \mathbf{x}_i \hat{\boldsymbol{\beta}})^2}{V[Y_i]} = \sum_{i=1}^n w_i (Y_i - \mathbf{x}_i \hat{\boldsymbol{\beta}})^2$$

This sum is to be minimized using a least squares procedure as before, yielding the estimated regression coefficients.

Defining the matrix of observation weights as

$$\mathbf{W} = \begin{bmatrix} w_1 & 0 & \dots \\ 0 & \ddots & 0 \\ \vdots & 0 & w_n \end{bmatrix} = \begin{bmatrix} 1/\sigma_1^2 & 0 & \dots \\ 0 & \ddots & 0 \\ \vdots & 0 & 1/\sigma_n^2 \end{bmatrix}$$

the expression for the vector of estimated regression coefficients is now given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{Y}$$

Note that for Ordinary Least Squares, the weight matrix is proportional to the identity matrix, $\mathbf{W} \propto \mathbf{I}$, and the expression reduces to the usual.

The obvious problem with the Weighted Least Squares method is that usually the variances of the observations are not known, but must be estimated. There are, however, various approaches for overcoming this obstacle.

Firstly, it is sufficient for the method that the weights are *proportional* to the observation variances (not necessarily equal to them).

$$w_i \propto \frac{1}{\sigma_i^2}$$

This is because any proportionality constant will cancel out in the expression for $\hat{\beta}$.

This case is thus equivalent to knowing the ratios of the observation variances, σ_p^2/σ_q^2 , for any (p, q) .

These ratios – or “relative variances” can often be roughly estimated by inspection of a plot of estimated residuals against response variable observations.

Secondly, often the variances are varying with one of the predictor variables in a simple way that might be roughly estimated by inspection of a plot of estimated residuals against the relevant predictor variable observations. If, e.g., the relation can visually be estimated to be approximately a simple proportionality

$$\sigma_i^2 \propto x_{i1}$$

the weights can be defined to be reciprocal proportional to the relevant predictor variable observations

$$w_i \propto \frac{1}{x_{i1}}$$

Finally, if several observations are available for approximately the same combinations of predictor variables, the variances can be estimated by the usual sample variance estimation scheme.

21 Non-linear predictor variable terms in model

If the proposed UMLR model seems to be un-fitting, as may be indicated either by the hypothesis test for an overall significant regression model or by failing to reasonably fulfill the assumptions on the residuals, this may be because the model is too simple. Of course important predictor variables may be missing in the model altogether, but it is also possible that non-linear terms of the already included predictor variables may be missing. While the first issue is obviously application-dependent, the latter issue will be addressed in this section.

The non-linear terms of predictor variables may be of various types such as

- interaction terms between predictor variables
- polynomial terms
- more general non-linear terms
- combinations of these

The important constraint is here that the non-linearities are supposed to be only in the predictor variables and not in the unknown regression coefficients. In this way the basic UMLR model can easily be generalized to include such non-linear terms, while still be classified as a linear regression model, because it is linear in the coefficients. As will be seen in the following, this generalization requires only small modifications to the basic UMLR methodology.

As an example, suppose a UMLR model with two predictor variables, x_1 and x_2 , is allowed to include second order terms of the variables, x_1^2 and x_2^2 , as well as a second order interaction term, x_1x_2 . Thus the example model is

$$Y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_1^2 + \beta_4x_2^2 + \beta_5x_1x_2 + e$$

The error term is as usually assumed to be a random variable with zero mean and constant variance

$$E[e] = 0 \quad V[e] = \sigma^2$$

Note, that the equation, as just stated, is linear in the regression coefficients $\beta_0, \beta_1, \dots, \beta_5$.

The observations are then

$$Y_i = \beta_0 + \beta_1x_{i1} + \beta_2x_{i2} + \beta_3x_{i1}^2 + \beta_4x_{i2}^2 + \beta_5x_{i1}x_{i2} + e_i, \quad i = 1, 2, \dots, n$$

In this case it is convenient to use centered (or “mean-corrected”) predictor variable values, i.e.

$$\begin{aligned} x_{i1} &= X_{i1} - \bar{X}_1 \Rightarrow \bar{x}_1 = 0 \\ x_{i2} &= X_{i2} - \bar{X}_2 \Rightarrow \bar{x}_2 = 0 \end{aligned}$$

Here X_{i1}, X_{i2} are the original predictor variable values, $\bar{X}_1, \bar{X}_2$ are the sample means of these original values, and x_{i1}, x_{i2} are the centered versions of the predictor variable values. The reason for this simple transformation is that since the values x_{i1} and x_{i1}^2 (and likewise for x_{i2} and x_{i2}^2) will be highly correlated, some of the columns of the observation matrix $\mathbf{X}$ previously defined will also be highly correlated, giving numerical problems in the matrix inversion in the regression coefficients estimator equation $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$ due to a close singularity.

The non-linear terms in the example model can be handled by introducing a new set of predictor variables $z_1, z_2, \dots, z_5$ given by

$$\begin{aligned} z_1 &= x_1, \quad \bar{z}_1 = 0 \\ z_2 &= x_2, \quad \bar{z}_2 = 0 \\ z_3 &= x_1^2 \\ z_4 &= x_2^2 \\ z_5 &= x_1 x_2 \end{aligned}$$

Using the new predictor variables, the observations of the UMLR model with the non-linear terms included can then be put in the same form as for the basic UMLR model, i.e.

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 z_{i1} + \beta_2 z_{i2} + \beta_3 z_{i3} + \beta_4 z_{i4} + \beta_5 z_{i5} + e_i, \quad i = 1, 2, \dots, n \\ E[e_i] &= 0 \quad V[e_i] = \sigma^2 \end{aligned}$$

As before, to take advantage of compact matrix notation, is defined

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{Z} = \begin{bmatrix} 1 & z_{11} & \cdots & z_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & z_{n1} & \cdots & z_{nk} \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix}$$

In this way the whole set of n simultaneous observations of the response variable and the new predictor variables can be written compactly as

$$\mathbf{Y} = \mathbf{Z}\boldsymbol{\beta} + \mathbf{e}$$

The sum of squares of estimated residuals is then

$$SSE = \sum_{i=1}^n \hat{e}_i^2 = (\mathbf{Y} - \mathbf{Z}\hat{\boldsymbol{\beta}})^T (\mathbf{Y} - \mathbf{Z}\hat{\boldsymbol{\beta}})$$

This sum is to be minimized using the least squares procedure as before, yielding the estimated regression coefficients expressed in the new predictor variable observations

$$\hat{\boldsymbol{\beta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{Y}$$

Using the calculated vector of estimated regression coefficients, the estimated response variable is for the example at hand expressed in the centered predictor variables x_1, x_2 as

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1^2 + \hat{\beta}_4 x_2^2 + \hat{\beta}_5 x_1 x_2$$

If desired, this equation can alternatively be expressed in the original non-centered predictor variables X_1, X_2 by inserting the transformations

$$\begin{aligned} x_1 &= X_1 - \bar{X}_1 \\ x_2 &= X_2 - \bar{X}_2 \end{aligned}$$

This gives the alternative form

$$\hat{Y} = \hat{\beta}'_0 + \hat{\beta}'_1 X_1 + \hat{\beta}'_2 X_2 + \hat{\beta}'_3 X_1^2 + \hat{\beta}'_4 X_2^2 + \hat{\beta}'_5 X_1 X_2$$

where

$$\begin{aligned} \hat{\beta}'_0 &= \hat{\beta}_0 - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2 + \hat{\beta}_3 \bar{X}_1^2 + \hat{\beta}_4 \bar{X}_2^2 + \hat{\beta}_5 \bar{X}_1 \bar{X}_2 \\ \hat{\beta}'_1 &= \hat{\beta}_1 - 2\hat{\beta}_3 \bar{X}_1 - \hat{\beta}_5 \bar{X}_2 \\ \hat{\beta}'_2 &= \hat{\beta}_2 - 2\hat{\beta}_4 \bar{X}_2 - \hat{\beta}_5 \bar{X}_1 \\ \hat{\beta}'_3 &= \hat{\beta}_3 \\ \hat{\beta}'_4 &= \hat{\beta}_4 \\ \hat{\beta}'_5 &= \hat{\beta}_5 \end{aligned}$$

When non-linear predictor variable terms are allowed in the UMLR model, the interpretation of the regression coefficients is not as clear as for the basic UMLR model.

For the basic model with two predictor variables but without non-linear terms, i.e.

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

is seen

$$\begin{aligned} \hat{\beta}_1 &= \frac{\partial \hat{Y}}{\partial x_1} \\ \hat{\beta}_2 &= \frac{\partial \hat{Y}}{\partial x_2} \end{aligned}$$

Thus, $\hat{\beta}_1$ is here the estimated change in the response variable pr. unit change in the predictor variable x_1 , and likewise for $\hat{\beta}_2$ and x_2 .

For the above example UMLR model with two (centered) predictor variables allowing non-linear terms

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1^2 + \hat{\beta}_4 x_2^2 + \hat{\beta}_5 x_1 x_2$$

is seen, however,

$$\frac{\partial \hat{Y}}{\partial x_1} = \hat{\beta}_1 + 2\hat{\beta}_3 x_1 + \hat{\beta}_5 x_2$$

$$\frac{\partial \hat{Y}}{\partial x_2} = \hat{\beta}_2 + 2\hat{\beta}_4 x_2 + \hat{\beta}_5 x_1$$

Therefore, here the estimated change in response variable pr. unit change in predictor variable depends on several estimated regression coefficients as well as on the values of the (centered) predictor variables.

To end the discussion of non-linear terms in the UMLR model, a few concluding remarks will be given.

First, even though most of the discussion here were in the form of an example, it should be clear how the methodology could be used in an analogue way for other combinations of polynomial and interaction terms.

In the same way, other non-linear terms than polynomial and interaction terms can also be included in the model as long as the non-linearities are in the predictor variables and not in the regression coefficients.

Secondly, it should be noted that the previously discussed inferences on regression coefficients, such as expressed in the ANOVA table and the regression table, are still valid even though non-linear predictor variable terms are introduced in the model.

Thirdly, for obvious reasons, particular caution should be used when doing extrapolations in a model with non-linear terms. Likewise, for polynomial terms, the highest polynomial degree should be chosen as low as possible.

Finally, it might be stated, that even though adding polynomial, interaction and general non-linear terms gives increased flexibility in the model, there is also an increased risk of over-fitting the model to the observations, and perhaps also an increased difficulty in choosing the proper terms for the model.

PART E4: Extensions 4:

Multivariate regression model

22 Multivariate multiple regression model

In some applications it is relevant to perform a linear regression analysis for a number of scalar response variables based on the same set of predictor variables. As an example, the response variables may be a company's electricity consumption, water consumption and various raw materials consumption over some time period, and the predictor variables may be various production specific quantities such as number of produced units of different sorts as well as e.g. meteorological data for the given time period.

A situation like this can be handled by considering the response variables *one-at-a-time* and thereby using a multiple univariate multiple linear regression model. Under certain circumstances, however, the statistical model, and therefore also its use for e.g. prediction purposes, can be improved by considering the response variables *simultaneous* dependency on the common predictor variables, giving a multivariate multiple linear regression model. Both cases will be discussed in this section with emphasis on the latter.

For the first case, a Multiple UMLR model for a number of response variables $Y_1, Y_2, \dots, Y_m$ based on a common set of predictor variables $x_1, x_2, \dots, x_k$ can be expressed as

$$\begin{aligned} Y_1 &= \beta_{01} + \beta_{11}x_1 + \dots + \beta_{k1}x_k + e_1 = [1 \ x_1 \ \dots \ x_k][\beta_{01} \ \beta_{11} \ \dots \ \beta_{k1}]^T + e_1 \\ Y_2 &= \beta_{02} + \beta_{12}x_1 + \dots + \beta_{k2}x_k + e_2 = [1 \ x_1 \ \dots \ x_k][\beta_{02} \ \beta_{12} \ \dots \ \beta_{k2}]^T + e_2 \\ &\dots \\ Y_m &= \beta_{0m} + \beta_{1m}x_1 + \dots + \beta_{km}x_k + e_m = [1 \ x_1 \ \dots \ x_k][\beta_{0m} \ \beta_{1m} \ \dots \ \beta_{km}]^T + e_m \end{aligned}$$

Of course, the unknown regression coefficients $\beta_0, \beta_1, \dots, \beta_k$ are specific for each response variable as indicated by the extra subscript indices.

Analogue to the basic UMLR model, the error terms $e_1, e_2, \dots, e_m$ are assumed to be zero mean and with constant, however optionally different, variances

$$\begin{aligned} E[e_1] &= 0 & V[e_1] &= \sigma_1^2 \stackrel{\text{def}}{=} \sigma_{11} \\ E[e_2] &= 0 & V[e_2] &= \sigma_2^2 \stackrel{\text{def}}{=} \sigma_{22} \\ &\dots \\ E[e_m] &= 0 & V[e_m] &= \sigma_m^2 \stackrel{\text{def}}{=} \sigma_{mm} \end{aligned}$$

If the m error terms, and thereby the m response variable terms, are uncorrelated, i.e. if

$$\text{Cov}[e_{v_1}, e_{v_2}] = \text{Cov}[Y_{v_1}, Y_{v_2}] = 0, \quad \forall v_1 \neq v_2 \quad v_1, v_2 \in \{1, \dots, m\}$$

then the variance-covariance matrix for the error terms is diagonal

$$\Sigma_e = \Sigma_Y = \text{Cov}[\{e_1, \dots, e_m\}] \stackrel{\text{def}}{=} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1m} \\ \sigma_{21} & \sigma_{22} & & \sigma_{2m} \\ \vdots & \vdots & & \vdots \\ \sigma_{m1} & \sigma_{m2} & \dots & \sigma_{mm} \end{bmatrix} = \begin{bmatrix} \sigma_{11} & 0 & \dots & 0 \\ 0 & \sigma_{22} & & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \sigma_{mm} \end{bmatrix}$$

In this case the m regression equations above can be considered one-at-a-time independent of the others, thus giving a multiple UMLR model. Therefore this case presents no new problems compared to the basic UMLR model considered in all previous sections.

If, however, the error terms can not in advance be assumed to be uncorrelated, the regression model needs to be built as a true multivariate multiple linear regression model (MMLR model). Another way of putting this situation is that even though the observation vectors are independent from each other, the individual variables *within the same observation vector* are not independent, and therefore neither uncorrelated. This case requires quite a few modifications to the basic UMLR model, as will be sketched below.

The MMLR model can be stated as

$$[Y_1 \ Y_2 \ \dots \ Y_m] = [1 \ x_1 \ \dots \ x_k] \begin{bmatrix} \beta_{01} & \beta_{02} & \dots & \beta_{0m} \\ \beta_{11} & \beta_{12} & & \beta_{1m} \\ \vdots & \vdots & & \vdots \\ \beta_{k1} & \beta_{k2} & \dots & \beta_{km} \end{bmatrix} + [e_1 \ e_2 \ \dots \ e_m]$$

As can be seen, the response variables and error terms are now vectors as opposed to scalars in the UMLR model, while the regression coefficients are now a matrix as opposed to a vector in the UMLR model. The common predictor variables are still a vector.

The error terms are still assumed to be zero mean

$$E[\{e_1, e_2, \dots, e_m\}] = \mathbf{0} = [0 \ 0 \ \dots \ 0]$$

but now with a general, not necessarily diagonal, variance-covariance matrix

$$\text{Cov}[\{e_1, \dots, e_m\}] = \Sigma_e = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1m} \\ \sigma_{21} & \sigma_{22} & & \sigma_{2m} \\ \vdots & \vdots & & \vdots \\ \sigma_{m1} & \sigma_{m2} & \dots & \sigma_{mm} \end{bmatrix}$$

Next, to handle observations in the stated MMLR model in a concise way, below will systematically be used the following indexation

Predictor variable indexing	$j = 0, 1, \dots, k$
Response variable indexing	$v = 1, 2, \dots, m$
Observation indexing	$i = 1, 2, \dots, n$

Using these indexation conventions, the various observed and unknown quantities can be notated as the following

Predictor variable observation	x_{ij} where $x_{i0} \stackrel{\text{def}}{=} 1$
Response variable observation	Y_{iv}
Unknown response variable observation error	e_{iv}
Unknown regression coefficient	β_{jv}

Using these notation conventions, the n observations for *a single one* of the m response variables can be stated on exactly the same form as for the basic UMLR model (except for the extra subscript index, v , indicating which of the response variables is presently considered)

$$\begin{bmatrix} Y_{1v} \\ \vdots \\ Y_{nv} \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_{0v} \\ \vdots \\ \beta_{kv} \end{bmatrix} + \begin{bmatrix} e_{1v} \\ \vdots \\ e_{nv} \end{bmatrix}, \quad v \in \{1, \dots, m\}$$

Collecting the observations column vectors for the individual response variables and their related unknown regression coefficients and related unknown error terms into matrices, the complete MMLR model can then be stated as

$$\begin{bmatrix} Y_{11} & Y_{12} & \cdots & Y_{1m} \\ Y_{21} & Y_{22} & & Y_{2m} \\ \vdots & \vdots & & \vdots \\ Y_{n1} & Y_{n2} & \cdots & Y_{nm} \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & & x_{2k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_{01} & \beta_{02} & \cdots & \beta_{0m} \\ \beta_{11} & \beta_{12} & & \beta_{1m} \\ \vdots & \vdots & & \vdots \\ \beta_{k1} & \beta_{k2} & \cdots & \beta_{km} \end{bmatrix} + \begin{bmatrix} e_{11} & e_{12} & \cdots & e_{1m} \\ e_{21} & e_{22} & & e_{2m} \\ \vdots & \vdots & & \vdots \\ e_{n1} & e_{n2} & \cdots & e_{nm} \end{bmatrix}$$

or simply

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

Note that $\mathbf{X}$ is unchanged compared to the UMLR model, while the three other quantities in the simple equation are all now matrices instead of vectors.

To clarify the implications of the above stated error terms assumptions on the observations error term matrix, this matrix can be considered either a column of row vectors or a row of column vectors as visualized below

$$\mathbf{e} = \begin{bmatrix} e_{11} & e_{12} & \dots & e_{1m} \\ e_{21} & e_{22} & & e_{2m} \\ \vdots & \vdots & & \vdots \\ e_{n1} & e_{n2} & \dots & e_{nm} \end{bmatrix} = \{\leftarrow \mathbf{e}(i, :) \rightarrow\}_{i=1, \dots, n} = \left\{ \begin{matrix} \uparrow \\ \mathbf{e}(:, v) \\ \downarrow \end{matrix} \right\}_{v=1, \dots, m}$$

In the column-of-row-vectors representation (shown in the middle), each row, $\leftarrow \mathbf{e}(i, :) \rightarrow$, represent the m individual error terms within one observation, while in the row-of-column-vectors representation (shown to the right), each column represent the n error terms for a particular response variable Y_v .

First, all error terms are assumed zero mean

$$E[\mathbf{e}] = \mathbf{0}$$

Next, since the error terms from two observations vectors are assumed to be independent (except, of course, if the two vectors observation indices are identical)

$$Cov[\mathbf{e}(i_1, :), \mathbf{e}(i_2, :)]_{mm} = \delta_{i_1 i_2} \cdot \Sigma_e = \begin{cases} \Sigma_e = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1m} \\ \sigma_{21} & \sigma_{22} & & \sigma_{2m} \\ \vdots & \vdots & & \vdots \\ \sigma_{m1} & \sigma_{m2} & \dots & \sigma_{mm} \end{bmatrix} & i_1 = i_2 \\ \mathbf{0} & i_1 \neq i_2 \end{cases}$$

Moreover, since observations vectors are assumed independent, the only non-zero correlations between the n error terms for two particular response variables, Y_{v_1}, Y_{v_2} , are the correlations between the n pair-wise terms e_{iv_1}, e_{iv_2} , giving

$$Cov[\mathbf{e}(:, v_1), \mathbf{e}(:, v_2)]_{nn} = \sigma_{v_1 v_2} \cdot \mathbf{I}_{nn} = \begin{bmatrix} \sigma_{v_1 v_2} & 0 & \dots & 0 \\ 0 & \sigma_{v_1 v_2} & & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \sigma_{v_1 v_2} \end{bmatrix}$$

Alternatively, and perhaps easier interpretable, put in words: in the MMLR model, error terms within the same observations vector can be correlated, while different error terms from different observations vectors are uncorrelated.

Note that, as of yet, no assumptions have been made on the distribution of the error terms.

The least squares estimate of the matrix of unknown regression coefficients can be calculated as

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\beta}_{01} & \hat{\beta}_{02} & \dots & \hat{\beta}_{0m} \\ \hat{\beta}_{11} & \hat{\beta}_{12} & & \hat{\beta}_{1m} \\ \vdots & \vdots & & \vdots \\ \hat{\beta}_{k1} & \hat{\beta}_{k2} & \dots & \hat{\beta}_{km} \end{bmatrix} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

Note the similarity with the analogue UMLR equation even though $\mathbf{Y}$ and $\hat{\boldsymbol{\beta}}$ are now matrices instead of vectors.

The estimator is still unbiased

$$E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$$

and its variance-covariance matrix can also be expressed in a way much resembling the UMLR case

$$\text{Cov}[\hat{\boldsymbol{\beta}}(:, v_1), \hat{\boldsymbol{\beta}}(:, v_2)] = \sigma_{v_1 v_2} (\mathbf{X}^T \mathbf{X})^{-1}, \quad \forall v_1, v_2 \in \{1, \dots, m\}$$

Now, if further the error terms are assumed to be Gaussian distributed, then $\hat{\boldsymbol{\beta}}$ will be multivariate Gaussian distributed and thus uniquely determined by the above mean and variance-covariance matrix.

To address the estimator for the error terms variance-covariance matrix, $\hat{\boldsymbol{\Sigma}}_e$, first, the estimated response variable matrix is given by (again mimicking the UMLR case)

$$\hat{\mathbf{Y}} = \mathbf{X} \hat{\boldsymbol{\beta}}$$

Thus, the estimated residuals are

$$\hat{\mathbf{e}} = \mathbf{Y} - \hat{\mathbf{Y}} = [\mathbf{I} - \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T] \mathbf{Y}$$

with zero mean

$$E[\hat{\mathbf{e}}] = \mathbf{0}$$

Next, using

$$E[\hat{\mathbf{e}}^T \hat{\mathbf{e}}] = (n - k - 1) \cdot \boldsymbol{\Sigma}_e$$

and further again assuming the error terms to be Gaussian distributed, finally the (biased) maximum likelihood estimator for the variance-covariance matrix is given by

$$\hat{\Sigma}_{e,ML} = \frac{1}{n} \hat{\mathbf{e}}^T \hat{\mathbf{e}} \sim \frac{1}{n} W_m(n - k - 1, \Sigma_e)$$

where $W_m(n - k - 1, \Sigma_e)$ is short notation for a Wishart (matrix) distribution with $n - k - 1$ degrees of freedom and parameter Σ_e .

In the same way as for UMLR, this estimator may not be of primary interest in its own right, but is necessary for making inferences on regression coefficients, since it expresses the “uncertainties” on the estimates of these.

Having estimated β and Σ_e , parallel to the UMLR case, the next steps would then be the multivariate versions of model control, decomposition of the response variable variations total sum of squares (MANOVA), various inferences on regression coefficients such as hypothesis tests and confidence intervals, predictions and so on.

All these points follow more or less along the lines of the UMLR case, even though the complexity of the necessary theoretical results as well as of the numerical computations certainly increase somewhat. This will not be pursued any further here, since the intention was merely to present the multivariate version of the multiple linear regression model.

The bottom line is that, when appropriate due to internal correlations within observation vectors as clarified above, it is indeed possible to handle linear regression analyses for several response variables depending simultaneously on a common set of predictor variables, thereby improving the statistical model and its use.

References

Kutner, Nachtsheim, Neter & Li
Applied Linear Statistical Models, Fifth Edition
McGraw Hill, 2005
ISBN 9780071122214

Johnson & Wichern
Applied Multivariate Statistical Analysis, Sixth Edition
Pearson, 2007
ISBN 9780135143506

Walpole, Myers, Myers & Ye
Probability and Statistics for Engineers and Scientists, Eight Edition
Pearson, 2007
ISBN 9780132047678

Miller & Freund
Probability and Statistics for Engineers, Seventh Edition
Pearson, 2005
ISBN 9780131278400

Ross
Probability and Statistics for Engineers and Scientists, Fourth Edition
Academic Press, 2009
ISBN 9780123704832

Løvås
Statistikk for Universiteter og Høgskoler, Anden Utgave
Universitetsforlaget, 2008
ISBN 9788215002248

Knut Conradsen
En Introduktion til Statistik, Femte udgave
DTU, IMSOR, 1984